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Day Month Year

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Status of Thesis

Note: red = pre first edit, orange = pre second edit and beyond, green = done, blue = may not make it into the thesis, or it may be moved to the appendix.

- Chapter ??: Not started. Will write last/almost last
- Chapter ??: Apparently this is a good chapter...
- Chapter ??: Passed first drafting. Very Minor corrections to be made.
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's edits (see email from 29/09/2020 'quick question' for notes).
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's comments.
- Chapter ??: Passed first drafting. Still to apply Ken and bryan comments
- Chapter ??: Not written, vaguely planned
- Chapter 1: Significant amounts of thought has already gone into this.
- Chapter 1.11: Not started. Will write last/almost last

Chapter 1

Characterisation of InGaAs

Photodiodes for detecting 2 μ m Light

1.1 Good references

- 95% QE diode with cut-off at 2.4 μ m [159].
- semiconductor book [160]
- photodiode/optics book [161]

1.2 Introduction

The current generation of gravitational will be limited by the thermal noise due to brownian motion of the molecules that make up the mirror coatings. Since these thermally driven fluctuations are a limiting source of noise [162], proposed future gravitational wave detectors such as the Einstein Telescope [75] and LIGO Cosmic

Explorer [163] will use cryogenically cooled mirrors.

However, cryogenically cooled mirror coatings will not be compatible with the current laser wavelength, which is $1\text{ }\mu\text{m}$, and so future gravitational wave detectors may use either a $1.5\text{ }\mu\text{m}$ or $2\text{ }\mu\text{m}$ wavelength laser. Currently, $2\text{ }\mu\text{m}$ is of interest as there have been promising designs for low noise mirror coatings that are optimised for this wavelength.

Currently, as InGaAs is widely used semiconductor for detecting $1\text{--}1.6\text{ }\mu\text{m}$ light, suitable InGaAs photodiodes were available for detecting the gravitational-wave-signal-containing laser beam that emerges from the OMC. These photodiodes are low noise, linear, can detect many tens of milliwatts of light, and have a high quantum efficiency. Additionally, InGaAs is a popular material to use as it can be operated at room temperature.

As the current InGaAs photodiodes are not sensitive to $2\text{ }\mu\text{m}$ light, ‘extended’ InGaAs photodiodes may be used instead. These diodes are extended in the sense that their cut-off wavelengths are above $2\text{ }\mu\text{m}$. To achieve this longer cut-off wavelength, extended InGaAs contains a higher percentage of indium than normal; however, this extra indium comes at the expense of the structural integrity of the lattice structure. This can cause the quality of the photodiode to be degraded. Previous experiments have been limited by the quantum efficiency of the extended InGaAs photodiode that was used [164,165]. To obtain low enough electronic noise, the authors of [164] used the photodiodes without a bias. A photodiode with a quantum efficiency of up to 95% has been reported.

Alternative materials that could be used include HgCdTe and InSb. HgCdTe has potential, however there are still technical challenges to overcome in creating diodes that work at???. The properties of an InSb detector with a high reported QE is reported in chapter ??, however this detector would not be suitable for use in a gravitational wave detector.

In this chapter, the relationship between quantum efficiency, reverse bias and noise in Extended InGaAs Photodiodes is explored. A selection of extended InGaAs photodiodes were tested with the aim of understanding general trends in the behaviour of extended InGaAs diodes. It was found that to get a high quantum efficiency and linearity, the diodes must be operated with a bias. However, if the diodes were used with a high bias then the dark noise of the diode would exceed the shot noise level of the detected light. The dark noise could be reduced significantly by cooling the diodes and so future detectors may feature cooled read-out diodes. Generally speaking, the longer the cut-off wavelength, the worse the diodes noise was. Future work may include investigating the growth conditions for extended InGaAs photodiodes as it has been shown that the dark noise of a diode made from extended InGaAs can be reduced by orders of magnitudes by optimising this. However, at the time of writing (2020), commercially available InGaAs photodiodes for detecting 2 μm light that are suitable for use in a gravitational wave detector do not exist.

1.3 What is an Extended InGaAs Photodiode

A photodiode is a square law detector and so it produces a current that is proportional to the power of the light it is detecting. Usually, they are of the order 1 mm in

diameter. They are two-terminal devices, with an anode and a cathode. Photodiodes often follow a p-doped, intrinsic, n-doped (PIN) structure. PIN photodiodes are used in applications where there is milliwatts of light power.

To understand a PIN photodiode, it is useful to first consider a simpler PN photodiode. A pn junction consists of two types of semiconductor: p-doped and n-doped. In the P-doped region, holes (the majority charge carriers) are free to move whereas in the n-doped region electrons (the majority charge carriers) are free to move.

At the interface between the p-doped and n-doped region, majority carriers from each side diffuse across it. Then, the minority charge carrier combines with a majority charge carrier, and so no net current will flow out of the device. When in equilibrium, there are no free carriers in the p and n regions, and so these regions become charged. Due to the diffusion of majority charge carriers across the interface, there is an electric field between the p-doped and n-doped region. The region of the device containing no free carriers is called the depletion region.

When a photon is incident on a pn junction, it will create an electron-hole pair. If this photon is absorbed within the depletion region of the diode, then due to the electric field, the electron and hole will be separated. These electrons/holes will generate a current in the circuit connected to the diode. However, if a photon is absorbed outside the depletion region, then the electron-hole pair will not be separated and will recombine. These photons will not contribute to the current in the circuit.

To widen the depletion region, an external voltage can be used. When *reverse bias* is applied to a photodiode – **that is a positive voltage across the cathode and anode** –, there is a larger electric field difference at the interface between the p-doped and n-doped region. This increases the volume of the photodiode results in absorbed photons contributing to the photocurrent.

To further increase the width of the depletion region, a layer of intrinsic semiconductor can be used to separate the p and n-doped region. An intrinsic semiconductor has no charge as it has an equal number of excited electrons and holes. An electric field will be present in a larger volume of material and so more photons will be converted into electrons which leave the photodiode diode.

InGaAs is a widely used semiconductor for making photodiodes. Typically, the indium content is around 53%, sometimes this is made clear by writing $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$, so that the InGaAs can be lattice matched to InP substrate on which the photodiode is grown on. At this level of indium doping, the InGaAs photodiode will be suffer from few mechanical defects and so the diodes created with it will be of high quality. Typically they are sensitive up to wavelengths around 1.6 μm .

Longer wavelength sensitivity can be achieved by increasing the amount of indium doping. These photodiodes are often called ‘extended’ and can be sensitive for wavelengths up to 2.6 μm . One of the main issues faced when creating an extended InGaAs photodiode is the lattice mismatch between the InGaAs and the InP substrate on which its grown. Lattice mismatching can arise due to a strained crystal structure, and since the extra indium in an extended InGaAs photodiode causes this [166] it is hard to fabricate a low noise extended InGaAs photodiode [167].

Photodiode	Peak quantum efficiency (%)	Cut-off wavelength ^a (μm)	Diameter (mm)
FD10D	80 ^b	2.5	1
G12183-010k	77 ^b	2.5	1
G12183-005k	77 ^b	2.5	0.5
IG26x500S4i	86 ^c	2.5	0.5
IG24x500S4i	84 ^c	2.4	0.5
G12182-005k	84 ^d	2.1	0.5
G12182-010k	84 ^d	2.1	1.0

^a λ_{EM} for $\eta = 50\%$ ^b $\lambda_{\text{EM}} = 1.9\mu\text{m}$ ^c $\lambda_{\text{EM}} = 2.0\mu\text{m}$ ^d $\lambda_{\text{EM}} = 1.6\mu\text{m}$

Table 1.1: Photodiodes tested

Above 53% indium, a buffer layer of InGaAs with less indium in between the absorbing layer and the substrate can be used to mitigate the lattice mismatch slightly, but the dislocations still give rise to trap states which cause the generation and recombination of charge carriers [168]. Lattice mismatching is the source of the large amounts of dark current found in extended InGaAs as well as the $1/f$ noise.

similar circuit model?

1.4 What is Needed from the Read-out Photodiode in a Gravitational Wave detector

The photodiodes used in the readout of a gravitational wave interferometer should have a quantum efficiency of more than 99%. The reason for having such a high quantum efficiency requirement is that to achieve the baseline level of ~ 10 dB of squeezed light that future gravitational wave detectors will use, losses in the interferometer have to be extremely low as losses degrade the amount of squeezing achieved.

Photodiodes for gravitational wave detectors should have low $1/f$ dark noise so that they are not limited in the detection band by electronic noise. Third generation ground based gravitational wave detectors will be quantum noise limited down to about 10 Hz [169], so the $1/f$ noise must be below the shot noise of the photo-current above this frequency.

The $1/f$ noise could be avoided by using a heterodyne detection, however the shot noise would increase if the mixing of the modulation and demodulation waveforms was not time independent [133] and this limits the choice of waveforms. These schemes recover increase the SNR by increasing the signal, however the use of squeezing means that this approach would still not be sufficient as the noise would still be enhanced compared to a homodyne readout. However, it has been proposed that the increase in noise associated with heterodyne readout could be avoided by the use of a broadband squeezer [170].

Current gravitational wave detectors operate in the 1-10 mW range at the output [49] to ensure that the junk light from other modes is a small portion of the total signal. Thus, any new photodiode should be able to produce ~ 10 mA of photo-current.

The properties were investigated in a selection commercially available photodiodes. The photodiodes that were tested are listed in table ???. According to each of the diodes datasheets, none of them have 99% quantum efficiency; however, the quantum efficiency of a diode can be improved by increasing the bias voltage (see Section ??). Increasing the bias increases the dark current and dark noise of the photodiode and so for each diode a maximum reverse bias voltage was found. The

benchmark dark noise used was the photodiode’s dark noise at 10 Hz; the cut-off bias was determined by the voltage where the dark noise was equal to the shot noise of a 10 mA current. The bias voltage also affects the saturation limit of the photodiode. If a diode was found to have a low saturation threshold (~ 1 mA), then the diode is not fit for use in a gravitational wave detector.

1.5 Reverse Bias and the Quantum Efficiency of an Extended InGaAs Photodiode

To demonstrate the effect that reverse bias has on the quantum efficiency of a diode, the Thorlabs FD10D was used. The FD10D was exposed to a variety of beams so that saturation effects would be visible. Two styles of measurements were used: One where the reverse bias and the response of the diode were measured at DC, and a lock-in style measurement where the quantity $\frac{\Delta\eta}{\Delta V_B}$ was measured. The quantum efficiency should be independent of frequency if the diode is operating in a non-saturating regime, and so the two styles of measurements should yield the same outcome. To prove that the increase in signal is a result of increasing quantum efficiency and not some other mechanism, for instance an avalanche effect, the noise of the photocurrent was compared to the level expected for shot noise at a bias voltage that yielded an acceptable level of dark noise (see Section ?? for details.).

1.5.1 Method

To measure the photocurrent produced by the photodiode, a transimpedance amplifier that would be shot noise limited for 10 mA of photocurrent was used. This

circuit is shown in Appendix B.2. The photodiode circuit allowed for the reverse bias of the photodiode to be altered.

A 1550 nm wavelength laser (Pure Photonics PPL) was used to illuminate the photodiode (FD10D). The beam was split by a 50:50 beam splitter so that the light could be amplitude stabilised. The in-loop diode was a high performance InGaAs photodiode (C3061GH), and the out-of-loop diode was the diode under test. The laser was amplitude stabilised by actuating on the laser diode's current. Shot noise limited measurements could be made above 1 kHz. A half waveplate and polarising beam splitter were used to control the power level of the light incident on the FD10D, and a lens was used to set the size of the beam.

The FD10D was exposed to a variety of different beams to investigate the possibility that current density related phenomena could affect the results. This was done by changing the beam size and beam power. As the FD10D is 1 mm, the maximum beam width would be around 330 μm in order for less than 1% clipping losses. The powers used were in the milliwatt range as these types of beam powers are the ones that a gravitational wave detector would operated with. The methods in this section can only be used to measure relative increases in quantum efficiency, measurements of absolute quantum efficiency require [find ref ??](#) additional apparatus.

To measure the effect that changing the reverse bias voltage has on the quantum efficiency of a photodiode, two approaches were used. By measuring the increase in photocurrent at DC it is straight forward to determined the increase in quantum efficiency. Alternatively, one can perform a lock-in style measurement. If the quantum efficiency depends on the reverse bias of the photodiode, then modulating the

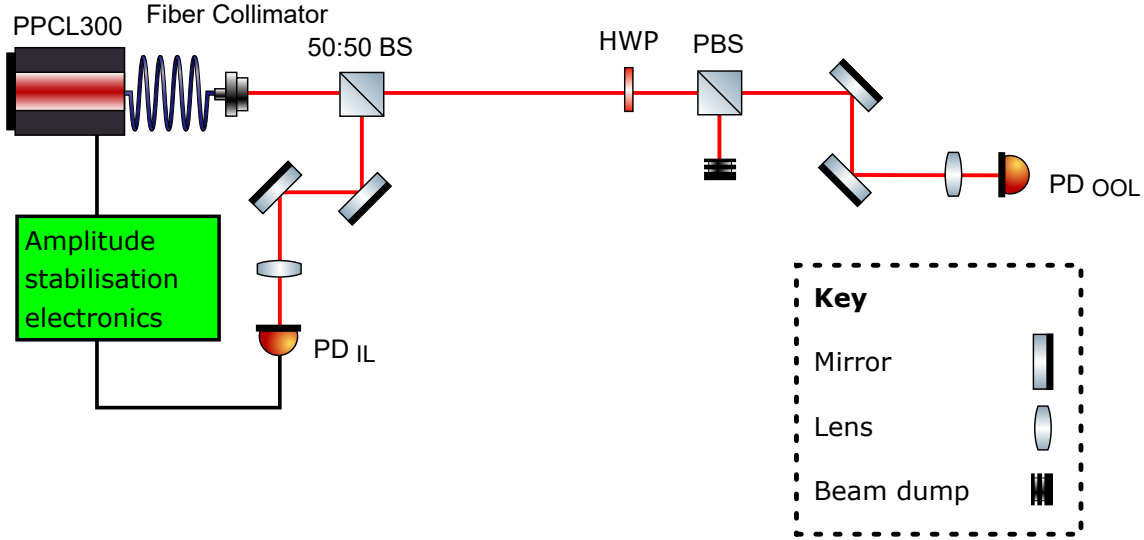


Figure 1.1: Optical layout used for measuring the effect that increasing the reverse bias of a photodiode has on its quantum efficiency and saturation limit. The out-of-loop photodiode, PD_{OOL} , was the photodiode under test. A 1550 nm laser (PPCL300) was used to test the optical properties of the photodiode. A combination of half wave plate (HWP) and polarised beam splitter (PBS) allowed for the power of the beam to be set and a lens was used to set the size of the beam on the photodiode under test. Amplitude stabilised light was needed to ensure the increase in photocurrent seen when the bias increased was due to an increase in quantum efficiency. The photodiode circuit (see Appendix B.2) allowed for shot noise limited measurements of currents around 10 mA and for the bias to be selected.

reverse bias, V_B at a frequency f will result in η modulating at f , and so the measurement will be of $\Delta\eta/\Delta V_B$. The input modulation $\Delta V_{\text{in}} = \Delta V_B$ and the output modulation $\Delta V_{\text{out}} \cos \theta = \Delta\eta V_{\text{TIA}}(V_B)$, where $V_{\text{TIA}}(V_B)$ is the DC signal from the photodiode at a reverse bias V_B and θ is the phase between the input and output modulation.

As 0.6 V reverse bias was the highest bias that gave an acceptable level of dark noise (see section ??), the noise of the photocurrent was measured at a region where the noise of the laser was limited by shot noise. This was done to ensure that the increase in photocurrent was due to a real increase in quantum efficiency and not due to another mechanism e.g. an avalanche process. The expected level of noise, n , of the photocurrent, in units of $A_{\text{rms}}/\sqrt{\text{Hz}}$, is

$$n = \sqrt{2qI_{\text{OOL}} + 2qI_{\text{IL}} + n_{\text{IL dark}}^2 + n_{\text{OOL dark}}^2 + n_{\text{loop}}^2 + n_{\text{analyser}}^2} \quad (1.1)$$

where q is the charge of the electron, $I_{\text{OOL/IL}}$ is the current through the out-of-loop and in-loop photodiodes, $n_{\text{OOL/IL dark}}$ is the dark noise of the out-of-loop diodes and in-loop photodiode, n_{loop} is the electronic noise of the loop and n_{analyser}^2 is the noise of the analyser. By tracking the height of an intensity modulation in the light for a series of powers up to 16 mW, the linearity of the diode could be investigated.

At this bias, the quantum efficiency was calibrated using two other power meters. The power incident onto the photodiode was measured and the power reflected by the window at small angles was measured.

1.5.2 Results

The effect of increasing the reverse bias on the quantum efficiency of the FD10D is shown in Figure 1.2. For all beams intensities that the photodiode was exposed to, the increase in quantum efficiency was $\sim 15\%$ as the reverse bias went from 0 V to 1.8 V, the stated maximum reverse bias in the datasheet of the FD10D. There may be a saturation related effect at the 1% level, beams with a lower intensity experienced a greater increase in η than beams with a higher intensity, however the quantum efficiency may not be uniform over the diode – near the edges η is greater. The difference between the DC and AC response could only be described by a non-linear process. These effects are small, and Figure 1.2 gives a clear incentive to increase the bias voltage.

Figure 1.3 shows the photocurrent measured has the noise level predicted by Equation 1.1, meaning that the increase in photocurrent at a reverse bias voltage of 0.6V is, without doubt, an increase in quantum efficiency. This measurement was performed with 4 different photocurrents, ranging from 7.5 mA to 15.5 mA, and over this range there was a 2% change in the ratio between the DC and AC measurement, this is shown in Figure 1.4. The SNR of this measurement is limited to the 1% level due to the difference between the calibration peak applied at 1111 Hz and the level of shot noise.

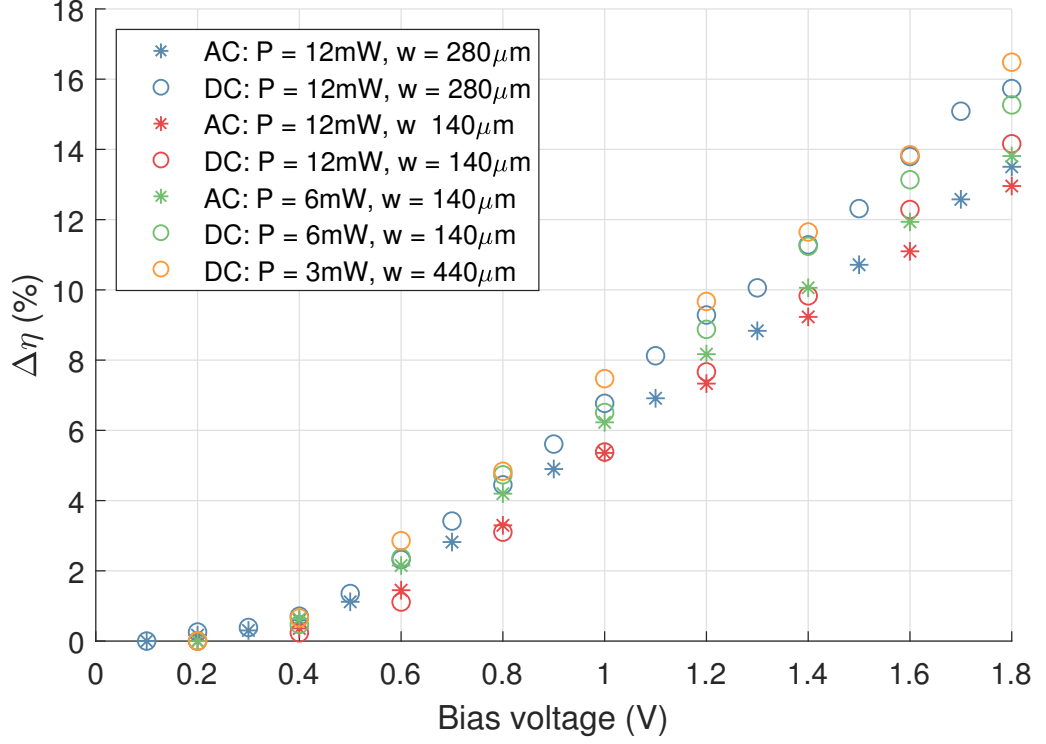


Figure 1.2: The FD10D's quantum efficiency depends on its reverse bias; the increase in quantum efficiency was at the $\sim 15\%$ level when the bias was increased to its maximum datasheet value. Two types of measurement were used to determine the increase in quantum efficiency as the bias was increased. The first was to measure the DC response and the corresponding results are shown as circles. The second method involved modulating the bias and measuring the resulting modulation in the photocurrent, i.e. an AC measurement, and these are shown by the stars. Each set of measurements was done with a different intensity by changing either the beam's width or power.

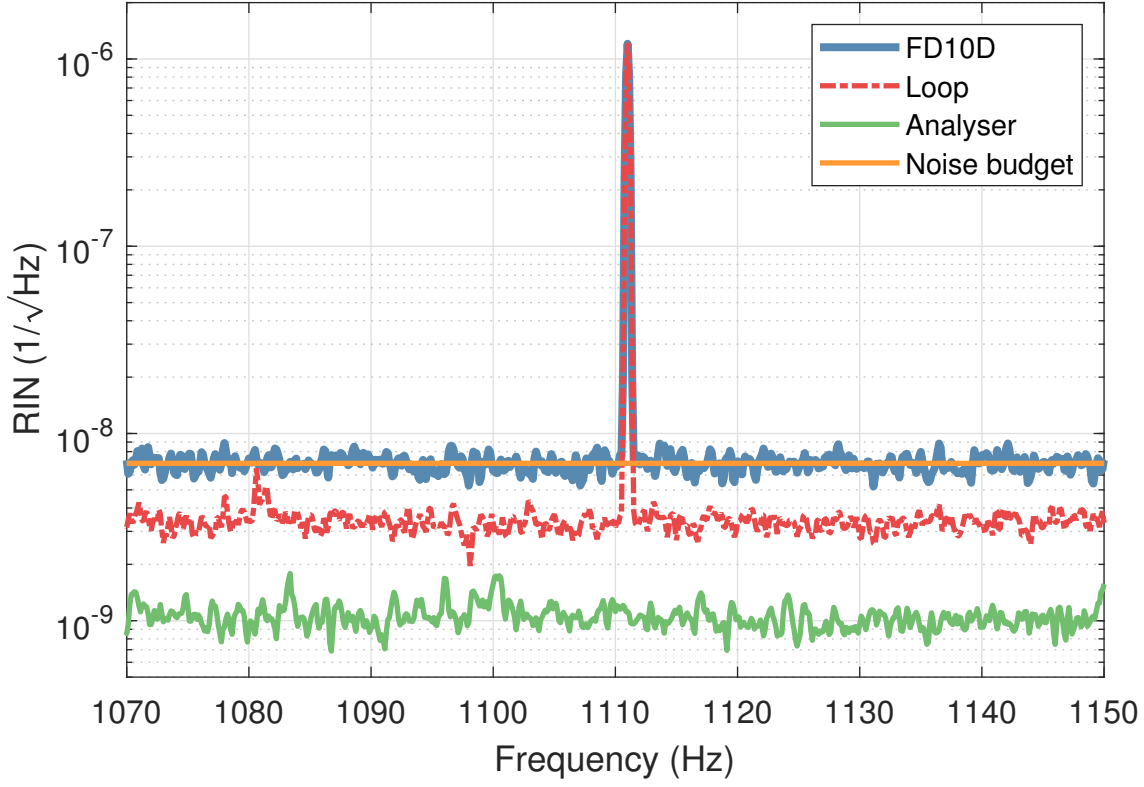


Figure 1.3: The bias of the FD10D was set to 0.6 V and it was exposed to light [find values](#). The noise of the photocurrent produced by the FD10D (blue line) corresponds to the Equation 1.1 (orange line) meaning that that the quantum efficiency of the FD10D did increase rather than the increase in photocurrent being the result of an avalanche process. The in-loop signal (red line) was used to cross-check the calibration of the measurement.

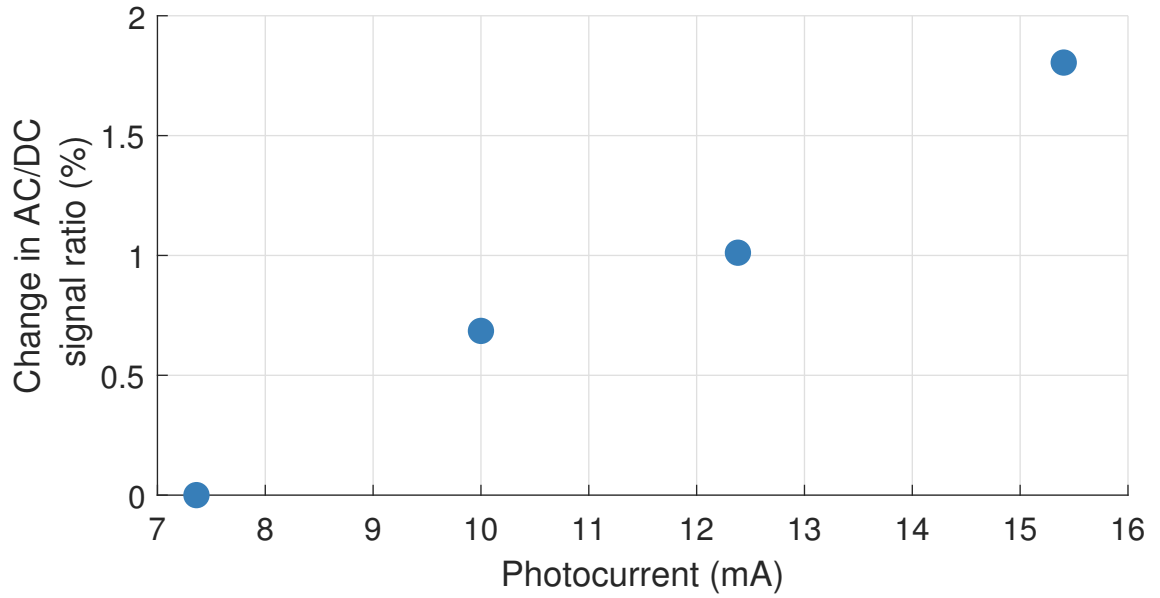


Figure 1.4: The bias of the FD10D was set to 0.6 V and it was exposed to a series of different power levels to check at what photocurrent the diode would saturate at. This was done by measuring the height of the calibration peak; if this peak decreased in size relative to the DC signal then this would indicate the AC signal was saturating. This peak could only be measured to within 1%, and so the FD10D showed no sign of saturation under these conditions.

1.6 Dark Current and Dark Noise in Extended InGaAs Photodiodes

The dark current and dark noise of extended InGaAs photodiodes has been a limiting factor in previous quantum optics experiments [164, 165]. The dark current of these photodiodes depends on their reverse bias. The larger the reverse bias, the higher the dark current. The dark current of extended InGaAs photodiodes is high because of the relatively large amounts of dislocations due to lattice mismatch between the InGaAs and the substrate that it is grown on.

There is a balance to strike between the quantum efficiency and dark noise. It is desirable to have as large a reverse bias as possible as this increases the quantum efficiency and saturation limit of the photodiode (see section ?? or figure ??). The dark noise of the photodiode should be below the shot noise level of the light that will be detected. As a bench mark, the dark noise of the photodiodes at 10 Hz is compared to the dark noise of 10 mA of photocurrent.

The noise of the current made by the photodiode can be modelled as

$$i_{\text{rms}} = \sqrt{2qi + \frac{4k_B T}{R_{\text{PD}}} + \left(\frac{i_t}{f^\beta}\right)^2}. \quad (1.2)$$

The first term under the square-root, $2qi$, is the shot noise associated with the DC current i . The second term, $\frac{4k_B T}{R_{\text{PD}}}$ is the Johnson-Nyquist noise due to the shunt resistance of the photodiode, R_{PD} ; this term can become significant when the shunt resistance is low as it is in some extended InGaAs photodiodes. The final term is Hooge's law of noise in semiconductors; i_t is the noise level and β is usually close to

0.5.

To measure the dark noise produced by the extended InGaAs photodiodes, a transimpedance amplifier with high gain was used. The photodiodes were placed in a light tight box while the measurement was being made.

1.6.1 Room Temperature

The dark noise as a function of bias for the photodiodes listed in table ?? was measured at room temperature ($\sim 20^\circ\text{C}$). The results for the FD10D are shown in Figure 1.5. The dark noise for the rest of the diodes can be found in Appendix A.1. Not all diodes had a simple noise spectrum, however usually the spectrum would fall as $1/f^{0.5}$.

The dark noise at 10 Hz and the dark current as a function of bias for each diode is shown in Figure 1.6. The noise at 10 Hz was either determined by reading the value from the corresponding graph in Appendix A.1 or, if more precision was required, by fitting a $1/f^\beta$ slope to the the data and inferring the noise from the fitted line.

Where there were duplicate diodes to choose from, the superior diode was selected for characterisation. There can be a significant amount of variation in quality within a batch of photodiodes. Figure 1.7 shows there was two orders of magnitude difference in noise spectrum of two ‘identical’ Hamamatsu photodiodes.

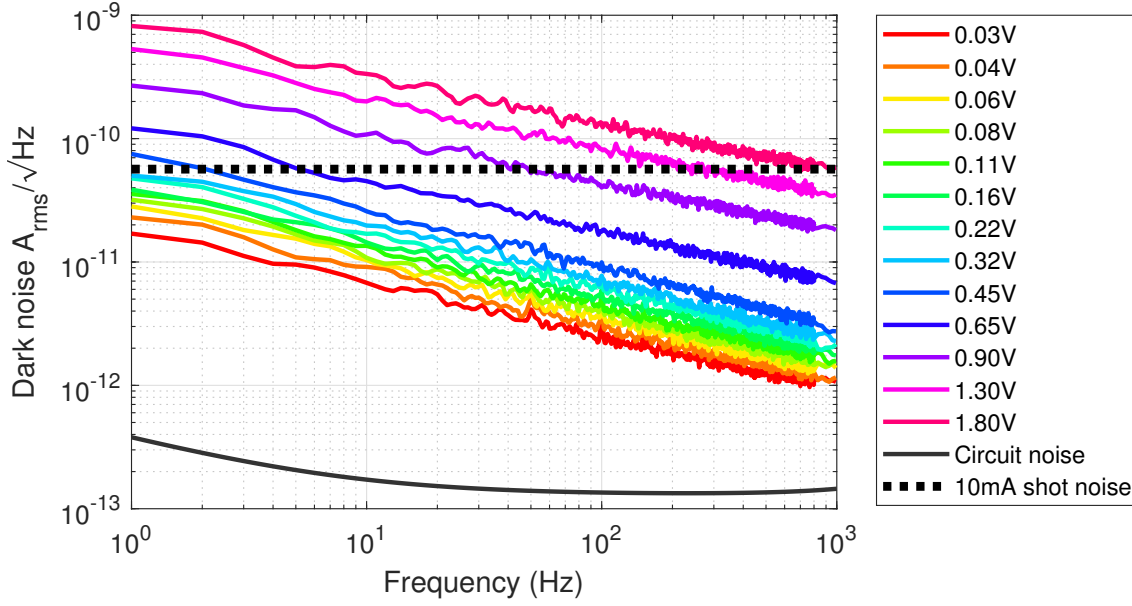


Figure 1.5: The dark noise of the FD10D increases as the reverse bias is increased. Above 0.6 V, the dark noise of the FD10D would limit the noise of a 10 mA current. As the noise spectral densities have no knees, the function i_t/f^β can be fitted to the it. See appendix A.1 for the dark noise spectra of other photodiodes. Below 0.9 V reverse bias, the spectra fall as $f^{-0.45}$ which is typical for ‘1/f’ noise. Above 0.9 V bias the spectra fall as $f^{-0.4}$. This data is shown along side the results for the other diodes tested in Figure 1.6.

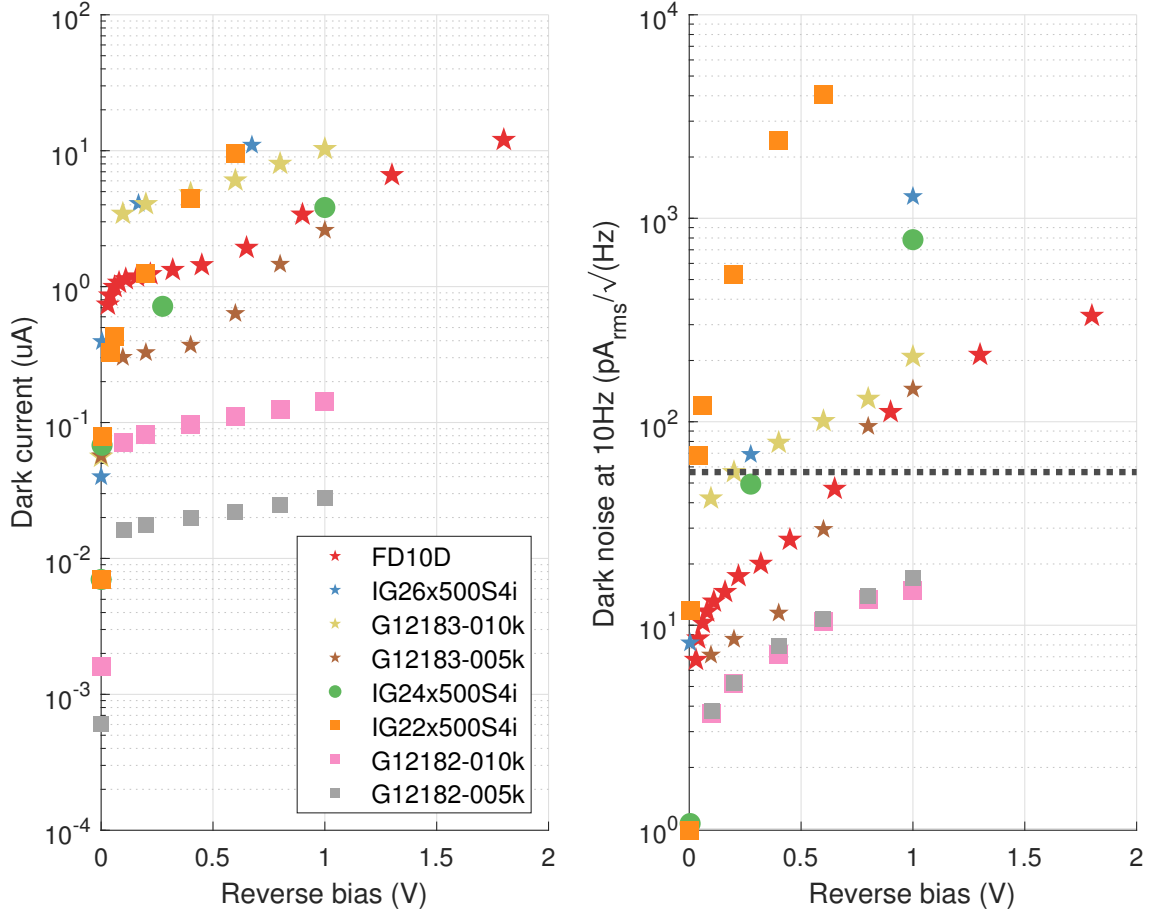


Figure 1.6: Left panel: the dark current for each diode was measured. Diodes of a similar cut-off wavelength share the same marker shape. The 1 mm photodiodes have larger markers than the 0.5 mm photodiodes. Right panel: the dark noise at 10 Hz for each diode. The dashed line indicates the level of shot noise for a 10 mA current.

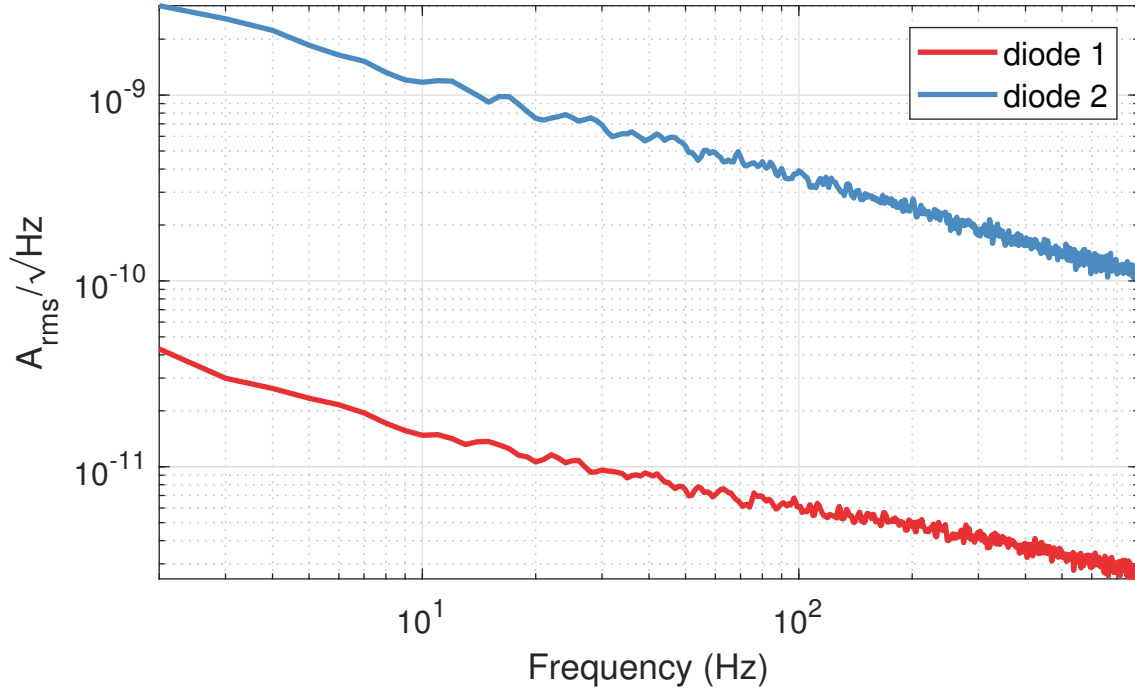


Figure 1.7: The difference in dark noise between two Hamamatsu photodiodes (g12182-010k) of the same kind under identical operating conditions (room temperature at 1 V bias). Diode 2 performs nearly 100 times better than diode 1.

1.6.2 Cooling Experiment

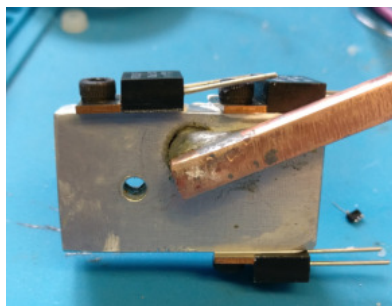
The typical low temperature damage threshold for extended InGaAs photodiodes is -40°C . To provide controlled cooling to the photodiode, the photodiode was glued (using Cryo-Varnish) to a block of aluminium, shown in figure 1.8. This block has a copper rod with a diameter of a few mm soldered to it, and this rod was inserted into a liquid nitrogen bath. The block of aluminium had power resistors to provide heating so that temperature of the diode could be controlled. A PID loop was implemented using a pt1000 temperature sensor and a multimeter (keithley 2000) configured to make a 4 wire resistance measurement. A labview program read this signal and performed calculated the error signal to be sent to a programmable power supply (Temna 72-2540) connected to the power resistors.

The dark noise and dark current for each diode in Table ?? was measured at the diode's maximum reverse bias specified in the data sheet. This is shown in figure 1.9 and the individual spectra are shown in Appendix A.2.

1.7 The Saturation Limit of Some Laser Components Extended InGaAs Photodiodes

The laser components diodes were found to saturate at low powers if they were biased with a low enough voltages such that their dark noise was below the noise target. The laser components photodiodes were exposed to a light signal of $\text{RIN} = 1.2\text{e-}6^1$. This could be cross-checked by measuring the light modulation on the

¹The peak cannot be expressed in spectral density units, however the height of the out-of-loop peak just needs to be compared to the in-loop sensor's measurement of the peak.



(a)



(b)

Figure 1.8: (a) The photodiode was affixed to this aluminium block. The power resistors provided heating to the block and the copper rod provided a thermal link between the liquid nitrogen dewar and the block. (b) A photodiode glued to the block such that a large proportion of the photodiode was in contact with it.

high performance in-loop photodiode. The power of the light was increased, and the decrease in the relative intensity of the AC modulation was recorded. This is shown in Figure 1.10. A decrease in AC signal indicates that the diode is saturating.

1.8 Excess Noise in the IG24x500S4i

The experiment explained in Section ?? showed that the noise of the photocurrent IG24x500S4i was higher than predicted by equation 1.1 by a factor of ~ 2 . The diode was not saturating during this measurement as the 1111 Hz peak has the right relative intensity.

To rule out bias noise causing this, a noise projection measurement was made. The transfer function between the bias at the anode of the photodiode and the output of the transimpedance amplifier was measured while 1 mA of photocurrent was

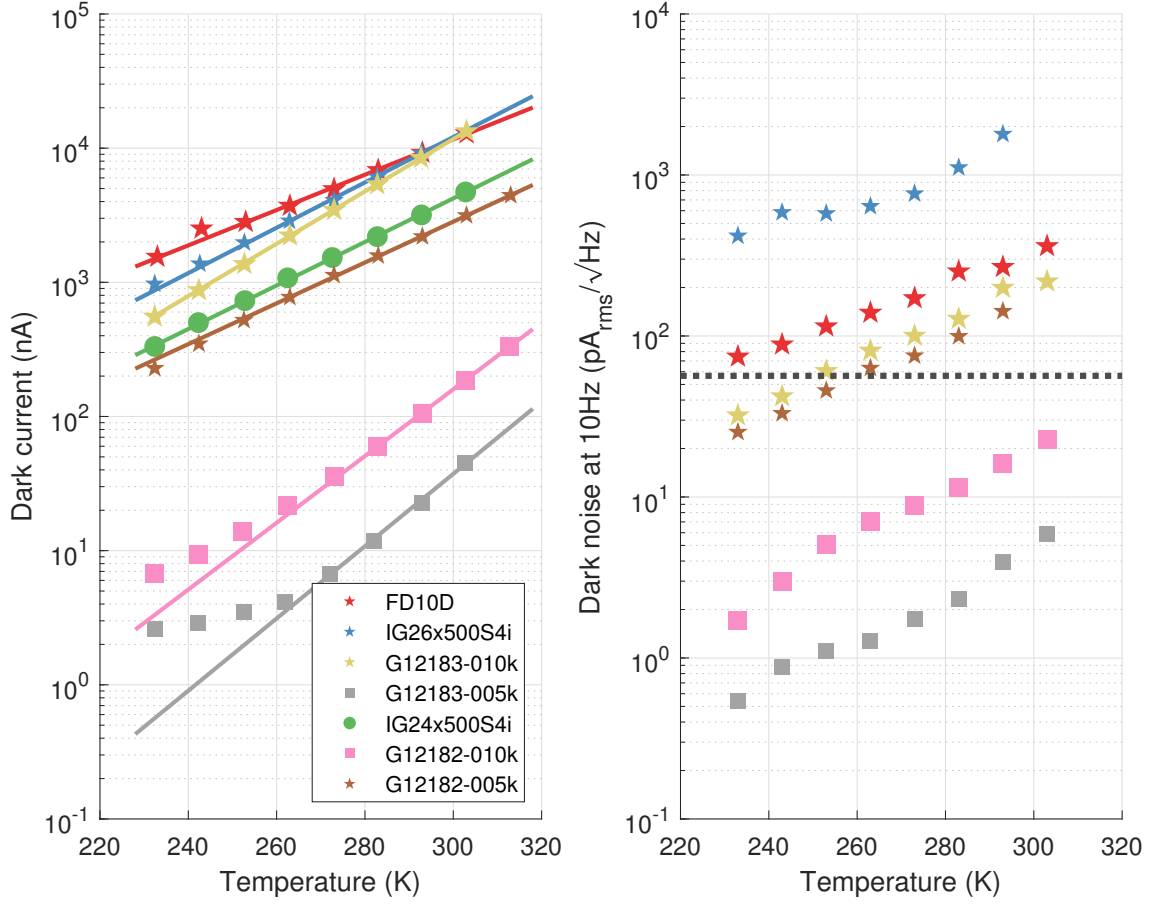


Figure 1.9: Left panel: the dark current of each photodiode at their maximum datasheet specified reverse bias as a function of temperature. The apparatus limited dark current measurements to a couple of nanoamps. Exponentials were fitted to the data. Diodes with similar cut-off wavelengths share marker shapes, and 1 mm diodes have larger markers than the 0.5 mm ones. Right panel: the dark noise of each diode was measured as a function of temperature. The black dashed line indicates the shot noise level for a 10 mA current. At 10 Hz, the measurements for the IG24x500S4i did not change much as the temperature changed, there was a knee around this frequency (see Figure A.13); the noise of the IG24x500S4i was $600 - 700 \text{ pA}/\sqrt{\text{Hz}}$ at 10 Hz.

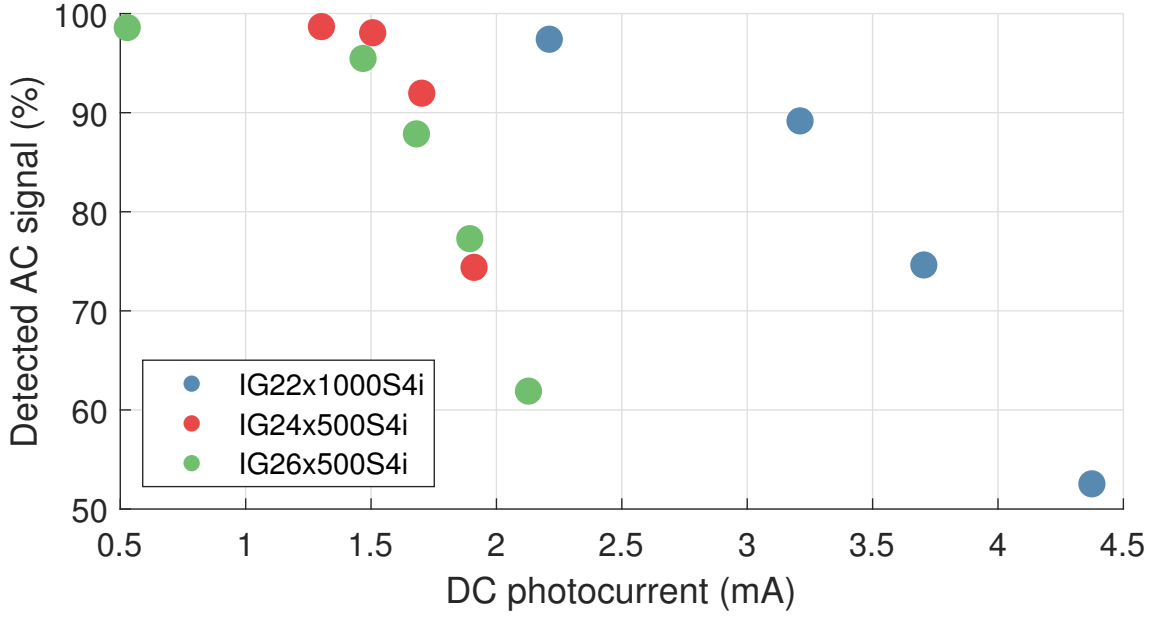


Figure 1.10: The saturation limit of three Laser Components photodiodes was measured. The photodiodes were biased so that their noise at 10 Hz was equal to the shot noise of a 10 mA current. The IG24x500S4i and IG26x500S4i were exposed to a beam whose 3σ width was 220 μm and the IG22x1000S4i was exposed to a beam whose 3σ width was 1000 μm . The size of the a light modulation whose relative size was known was tracked as the DC photocurrent was increased; the more saturated a photodiode is, the smaller the amount of signal was detected. These diodes had low saturations limits compared to the FD10D, which showed no signs of saturation up to 16.5 mA (see Figure ??).

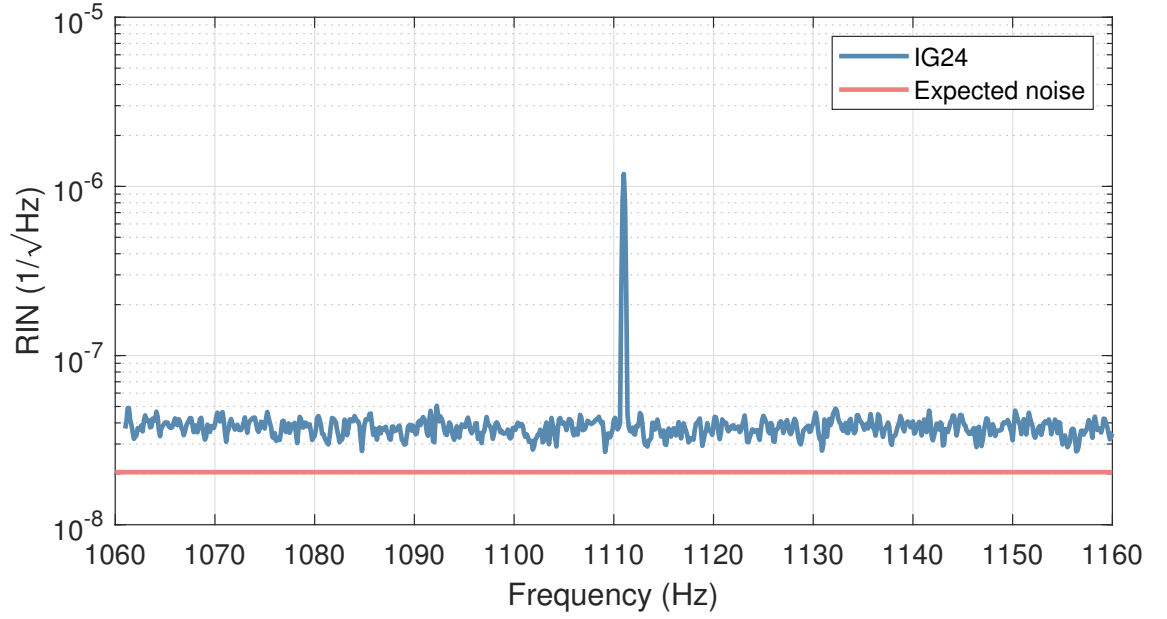


Figure 1.11: The noise of a 1.3 mA of photocurrent made by the IG24x500S4i (blue) exceeded the expected amount (red) based on Equation 1.1 by a factor of 1.75. This beam was not saturating the diode as the height of the 1111 Hz signal was the the right level. This excess noise was also found at 0.25 mA of photocurrent.

being generated by the diode. A calibration modulation was applied to the bias voltage, and the noise at the anode of the diode and the noise at the output of the transimpedance amplifier were simultaneously measured. Using this method, 42.5 dB of bias voltage noise was removed from the measurement; however, this did not remove the excess broadband noise. This is shown in figure 1.12.

The ideality factor of the IG24 was investigated. The equation of a diode is

$$I = I_p + I_s \left(\exp \left(-\frac{qV_b}{nk_B T} \right) - 1 \right), \quad (1.3)$$

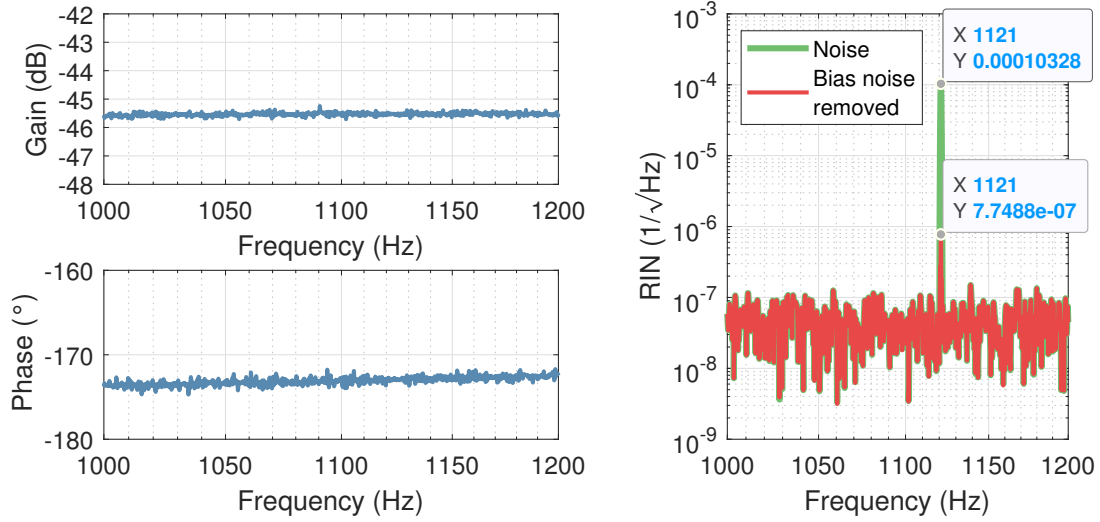


Figure 1.12: A noise projection measurement was performed to determine if the excess noise in the photocurrent produced by the IG24x500S4i (see Figure 1.11) was due to bias noise. The left hand panels show the bode plot of the transfer function between the photodiode's cathode and the output of the transimpedance amplifier when the photodiode was producing 1.3 mA of photocurrent. The right hand panel shows the effect of subtracting bias noise from the signal at the output of the transimpedance amplifier. The peak at 1121 Hz was a modulation applied to the photodiode's cathode; 42.5 dB of this signal was removed by using the noise projection technique which is in close agreement with the amount expected from the measured transfer function. However, the level of the broadband noise remained unchanged thus indicating that the excess noise witnessed in the photocurrent produced by the IG24x500S4i was not due to bias noise.

where I_p is the photocurrent, I_s is the saturation current of the diode, q is the charge of the electron, V_b is the bias of the diode, n is the ideality factor, k_B is Boltzmann's constant and T is the temperature of the diode. If n differs significantly from 1, then it indicates that additional behaviour that was not described in Section 1.3 occurring within the diode. The current flowing through the diode as a function of forward bias was measured and is shown in figure 1.13. The SciPy python function `scipy.optimize.curve_fit` was used to do an unweighted fit of the data to Equation 1.3. The diode was found to have $n \simeq 1$. To verify Equation 1.3, light was shone onto the diode. This is shown in figure 1.14.

1.9 InSb testing

An InSb detector was tested as its data sheet showed it had high quantum efficiency. The detector was characterised at by placing a a low emissivity material in front of the detector. The InSb is sensitive to wavelengths of up to be $5.3\mu\text{m}$. Based on the solid angle covered by the material as seen by the detector, the expected level of thermal radiation that would be detected could be calculated. This expected photocurrent was $5.5\mu\text{A}$. The DC current from the device was measured to be $5.3\mu\text{A}$. The results from this test are shown in Figure ???. This corresponds to the device having a quantum efficiency of $\sim 90\%$.

A reverse bias was applied to the device. This generated $1/f$ noise.... Full data is shown in Section A.3. The noise of the photocurrent also showed a broadband increase, this is shown in Figure 1.16.

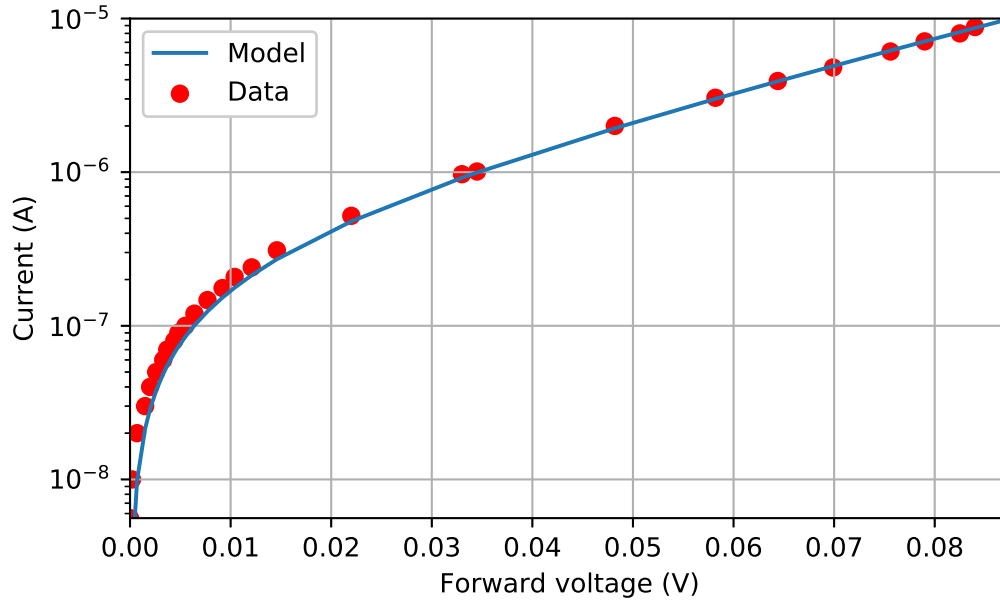


Figure 1.13: To investigate possible reasons why there was excess noise in the photocurrent produced by the IG24x500S4i, its ideality factor was measured (red points). Equation 1.3 was used to do an unweighed fit to the data ($I_0 = (360 \pm 10)$ nA and $n = 1.02 \pm 0.01$). The IG24x500S4i was found to be close to ideal, as n was close to 1, and so this does not provide clues to the source of excess noise in the photocurrent.

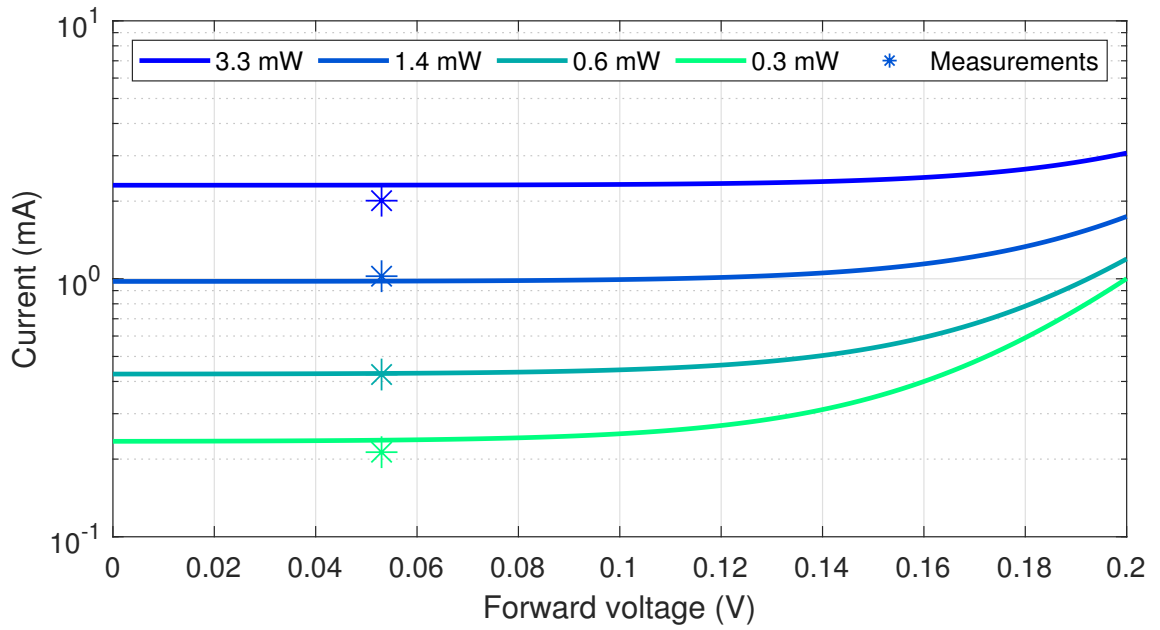


Figure 1.14: Different levels light levels were shone onto the photodiode and the current produced at 0.05 V reverse bias was recorded. These measurements are shown by the stars. Based on the model obtained from the data shown in Figure 1.13, the current flowing through the diode was predicted (indicated by the lines). The model and the measurements are in agreement with each other although the diode may have been saturating for the 3.3 mW measurement.

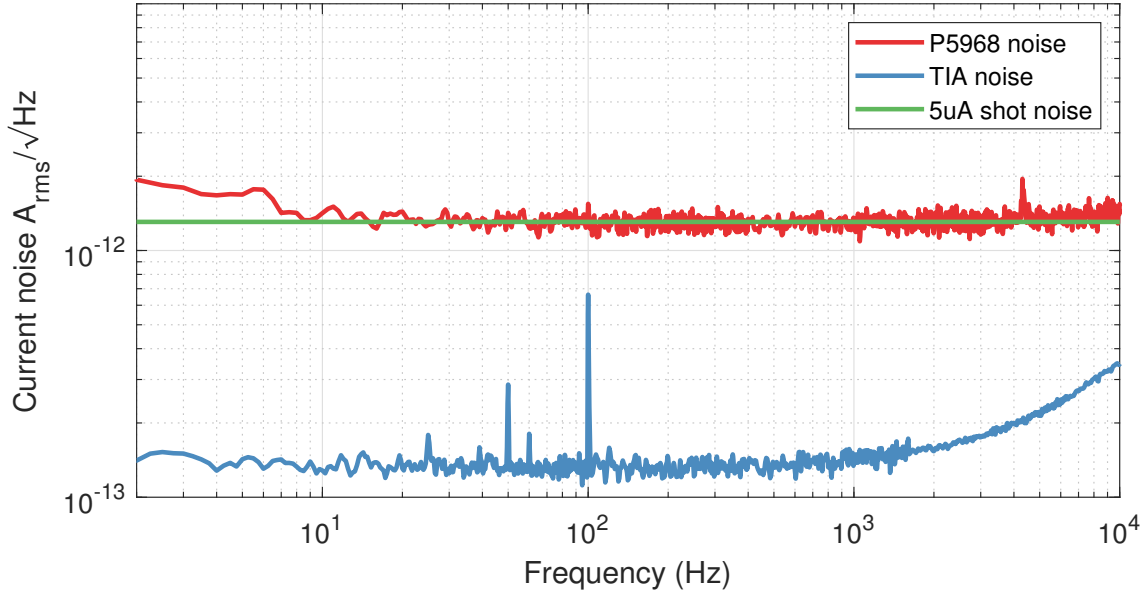


Figure 1.15: The P5968 had its cathode grounded and the noise of the photocurrent due to the blackbody radiation produced by room temperature aluminium was measured (red). The noise level of the photocurrent corresponds to shot noise (green). The noise of the transimpedance amplifier (blue) is well below the noise of photocurrents noise.

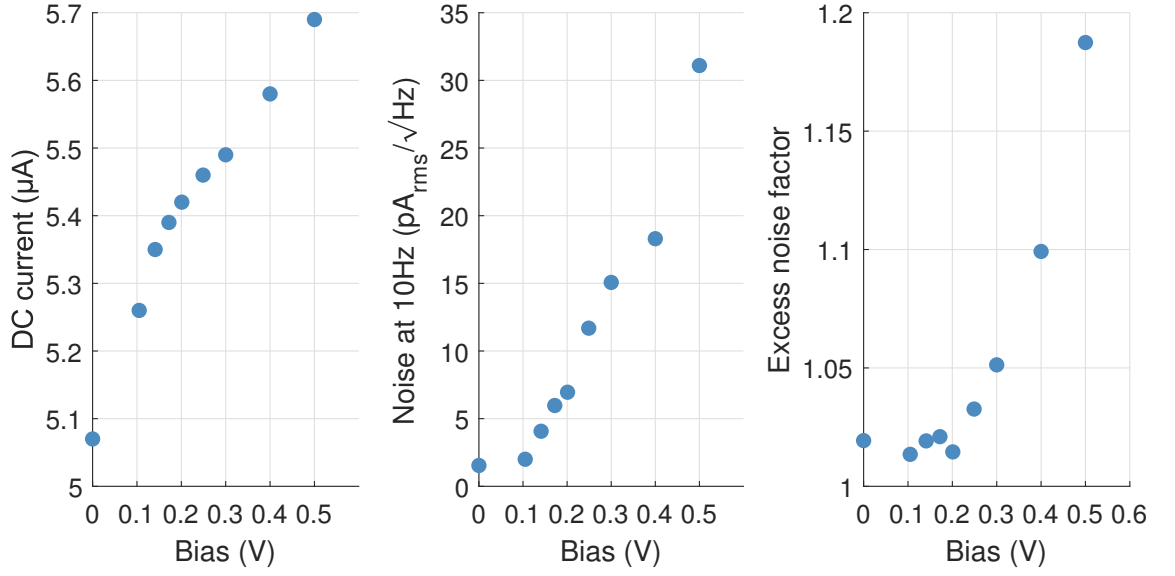


Figure 1.16: Left panel: the DC dark current of the P5968 as a function of bias. Middle panel: $1/f$ noise rose as the bias of the P5968 was increased (see Figure A.15). Right panel: At a frequency where $1/f$ was not present, the noise of the photocurrent was measured. The excess noise factor is the multiplicative factor that the shot noise corresponding to the DC photocurrent needs to be multiplied by get the noise level that was measured.

1.10 Discussion

Section 1.5 shows the effect that biasing an extended InGaAs photodiode can have on its quantum efficiency. The quantum efficiency of the FD10D was increased by 15% when its bias was set to its maximum value. This motivates using increasing the bias of a photodiode as the quantum efficiency can be greatly increased.

1.11 Conclusion

Appendix A

Extended InGaAs Diodes Noise Spectrums

A.1 Bias Section

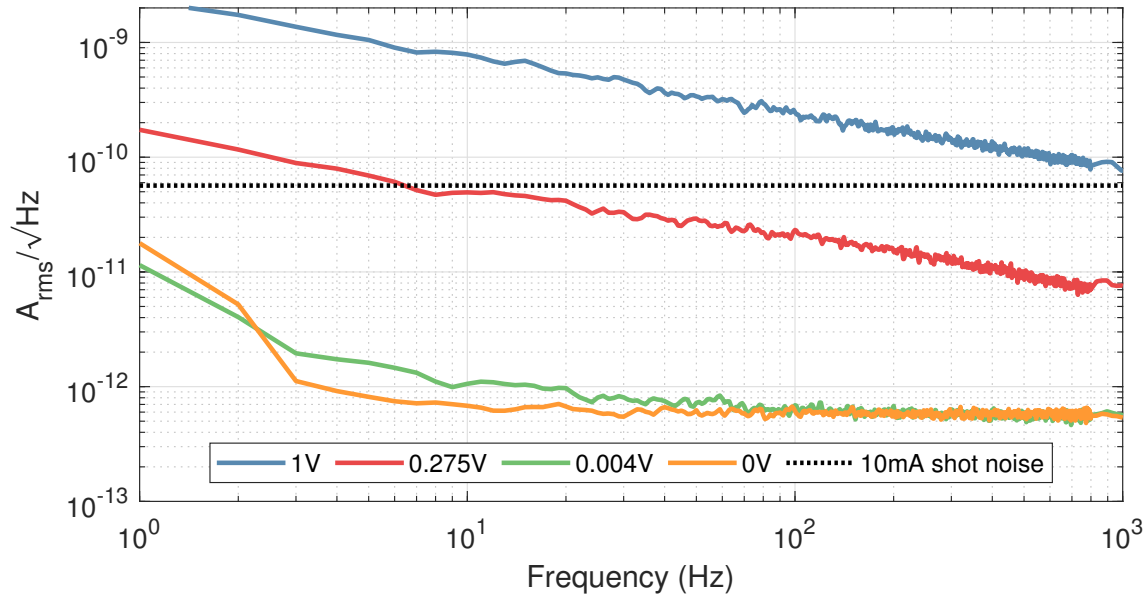


Figure A.1: IG24

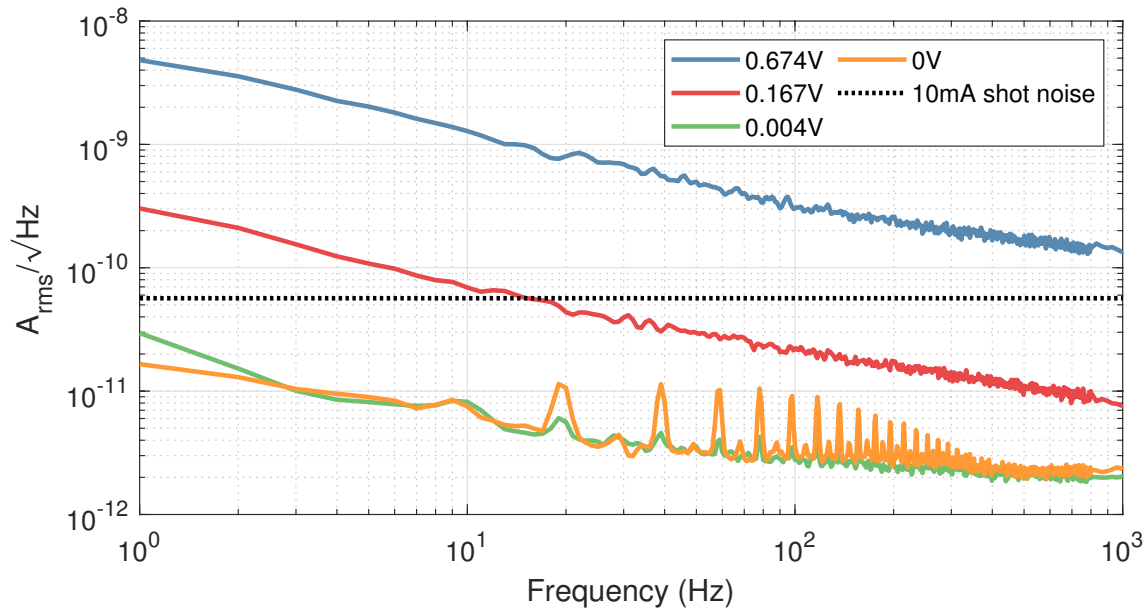


Figure A.2: IG26

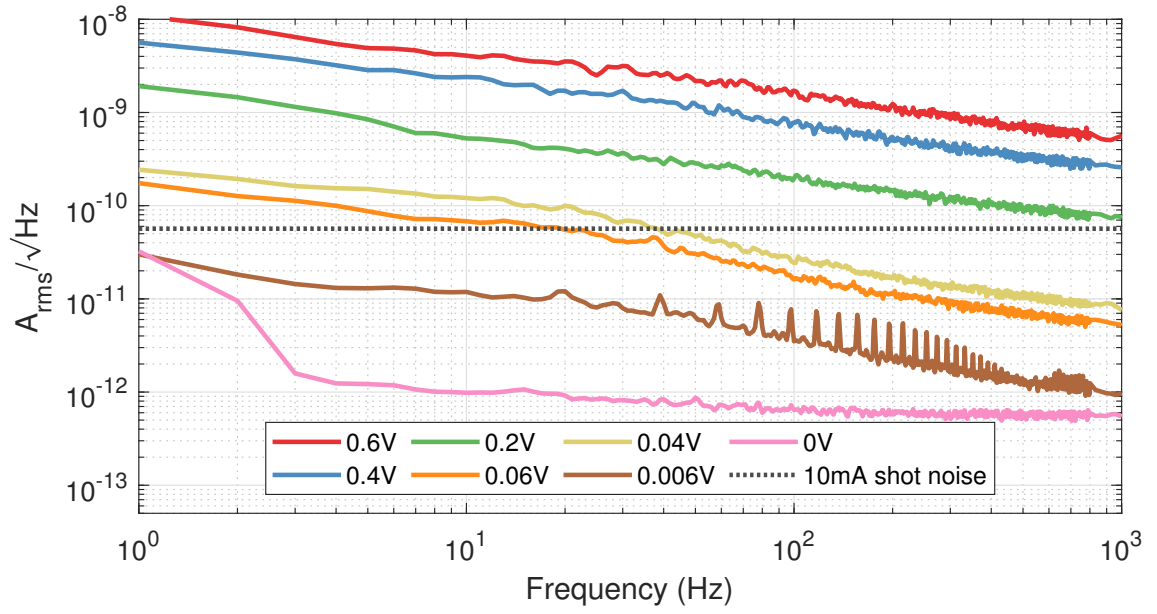


Figure A.3: IG22

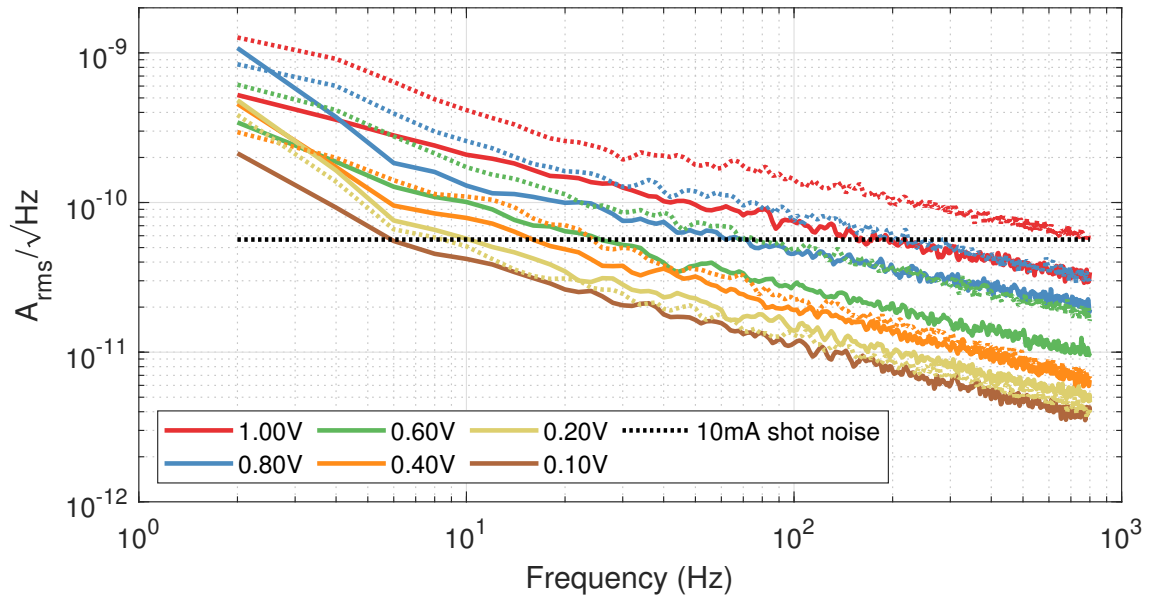


Figure A.4: G12183-010k

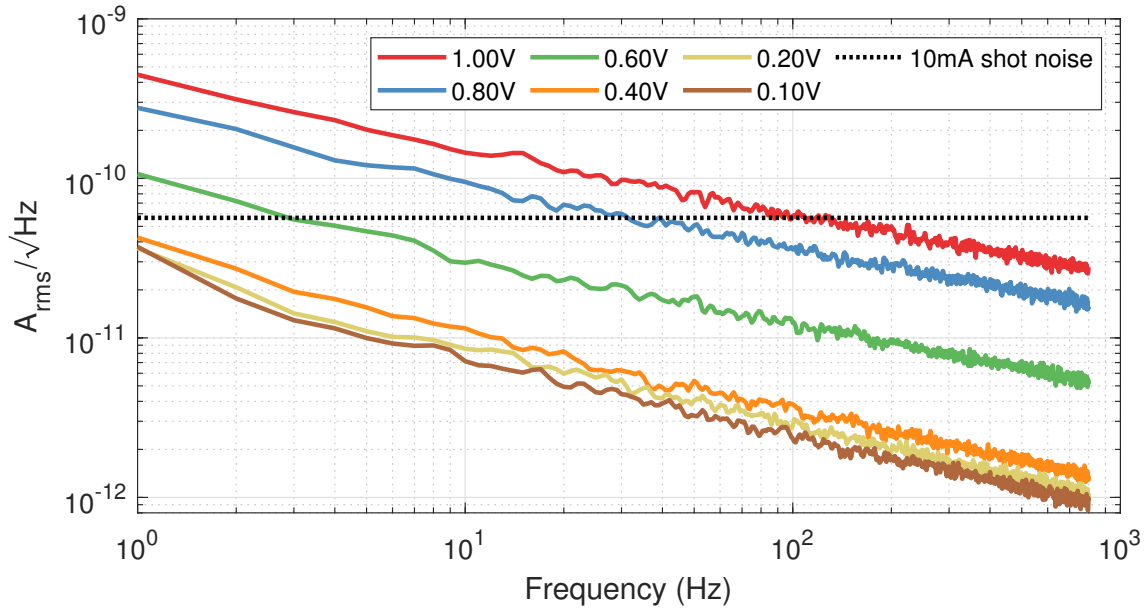


Figure A.5: G12183-005k

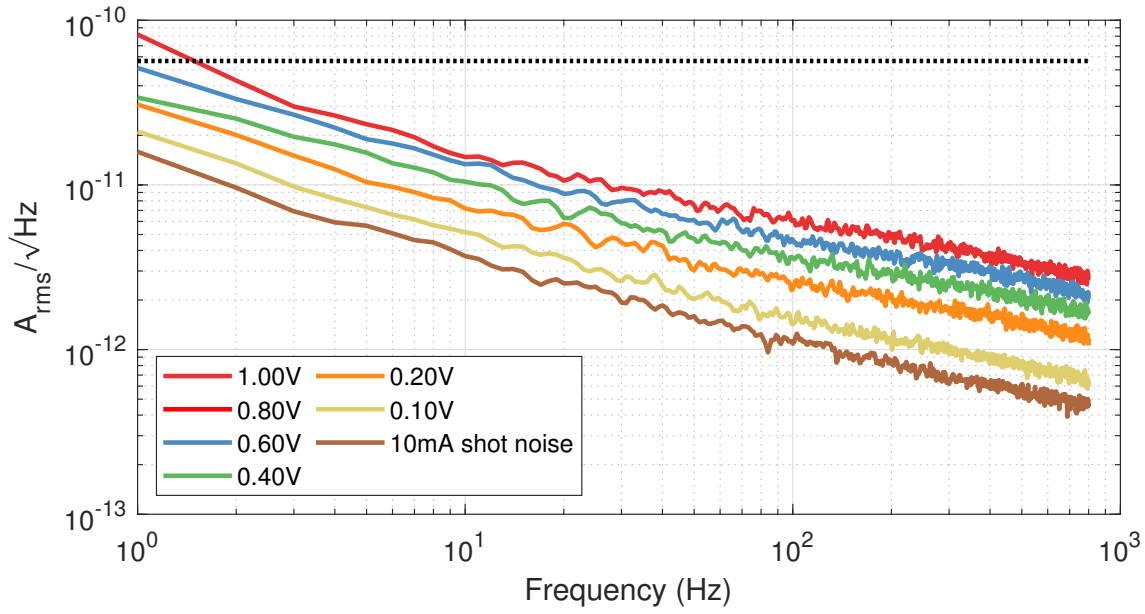


Figure A.6: G12182-010k

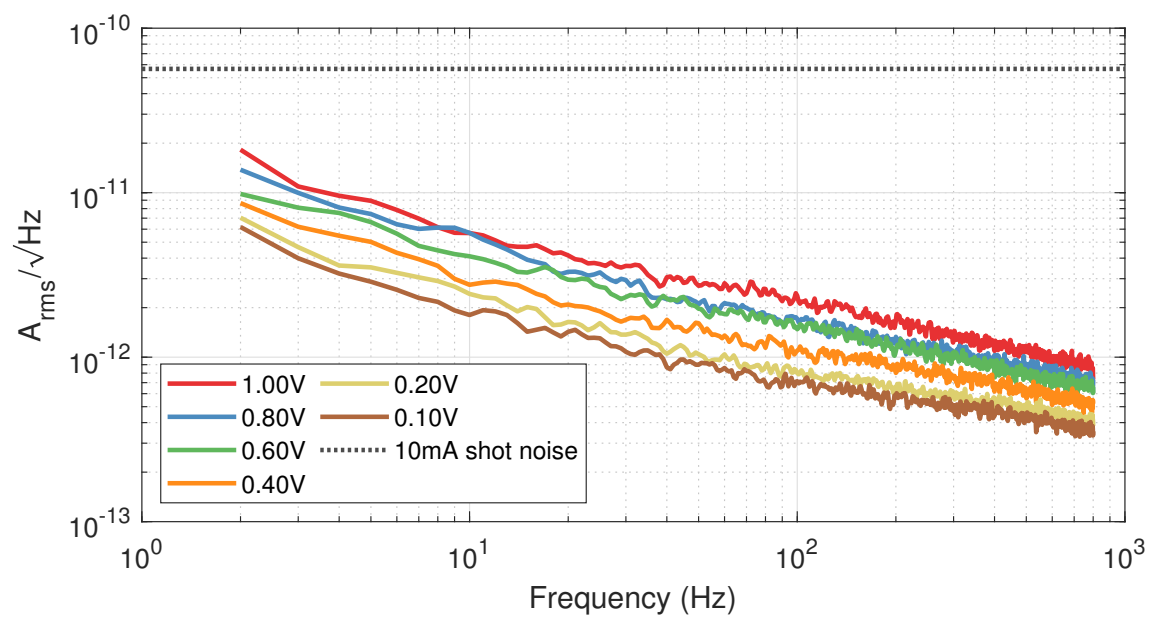


Figure A.7: G12182-005k

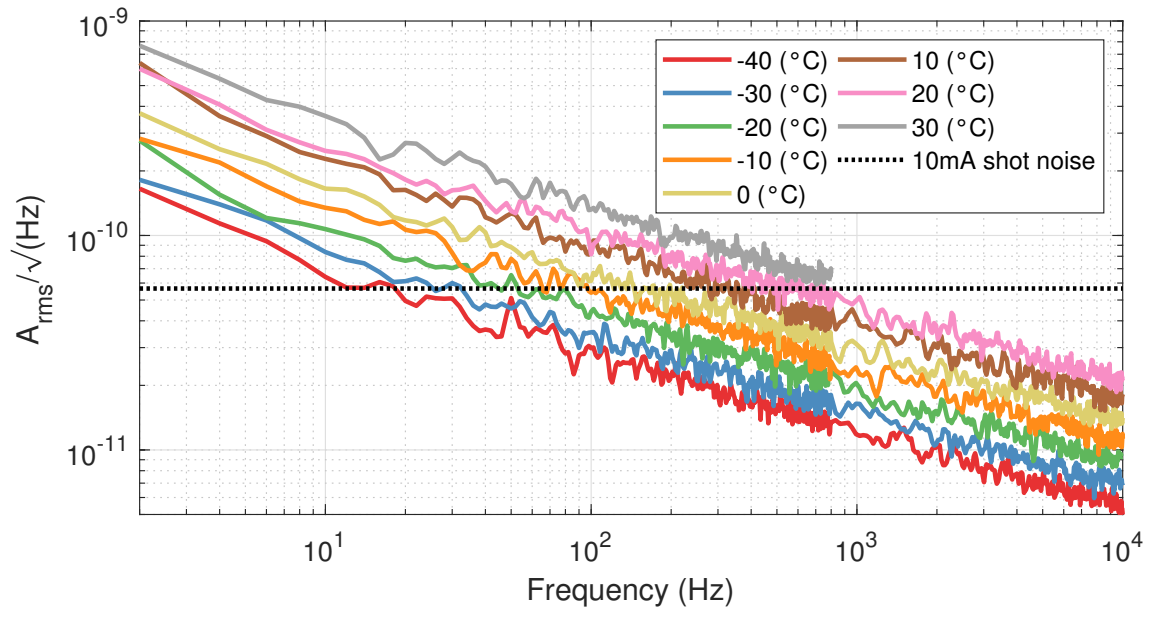


Figure A.8: FD10D 1.8V reverse bias

A.2 Temperature Section

A.3 InSb

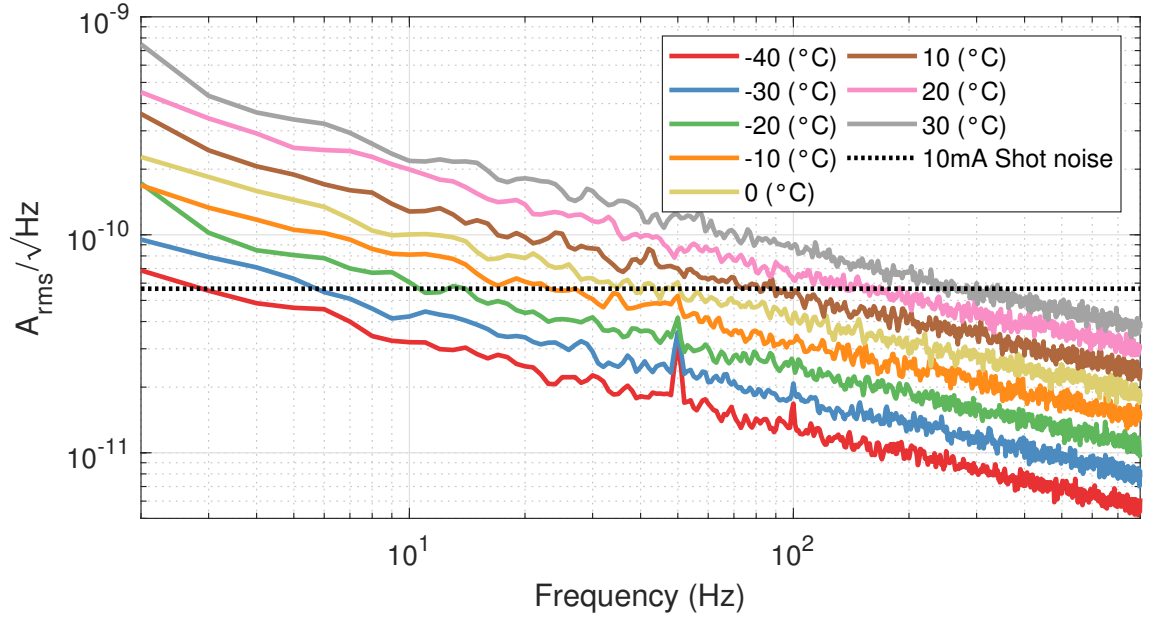


Figure A.9: G12183-010k 1.0V reverse bias

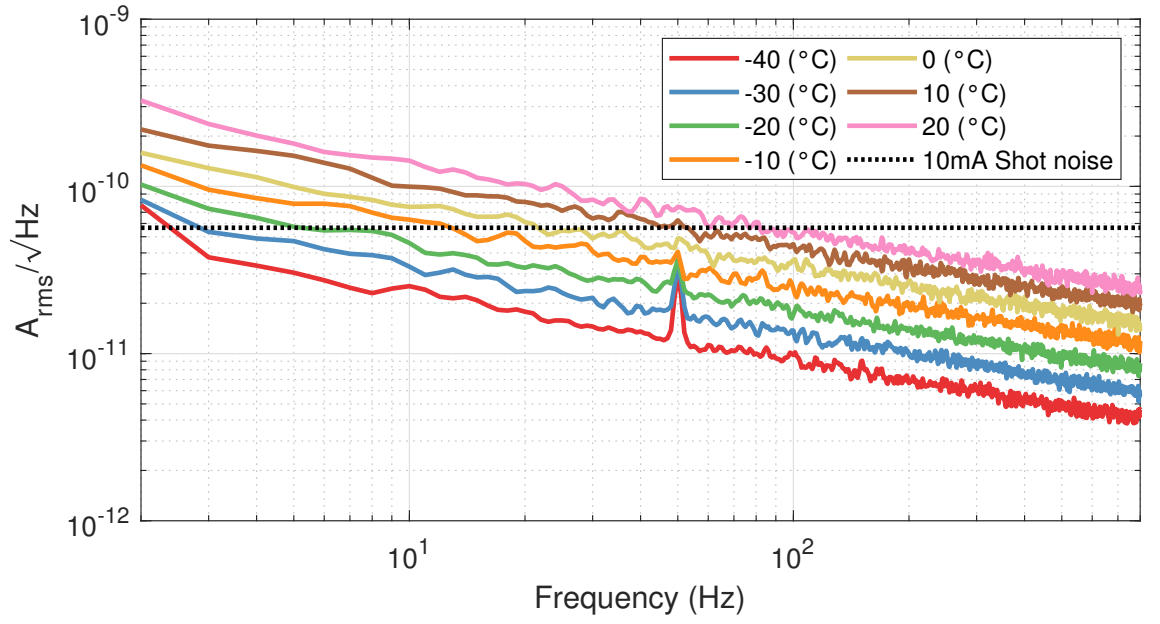


Figure A.10: G12183-005k 1.0V reverse bias

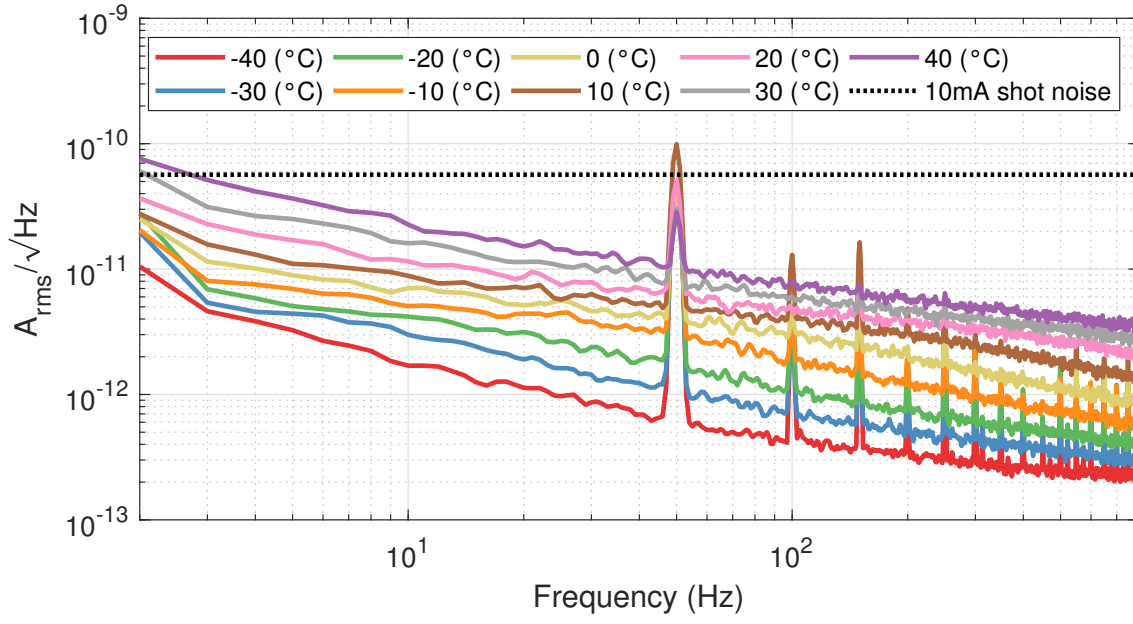


Figure A.11: G12182-010k 1.0V reverse bias

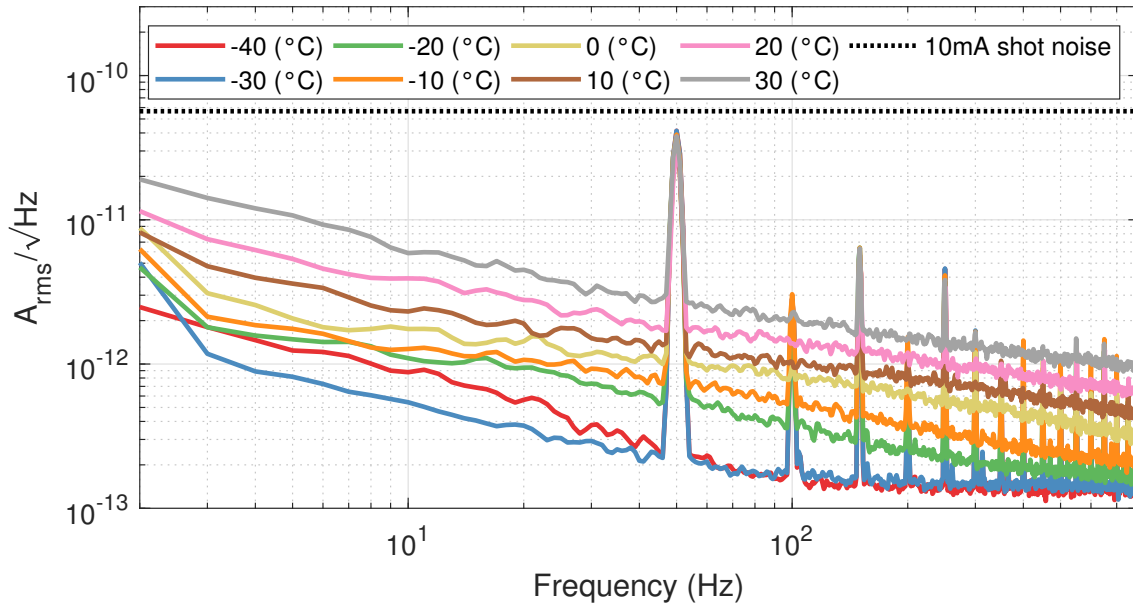


Figure A.12: G12182-005k 1.0V reverse bias

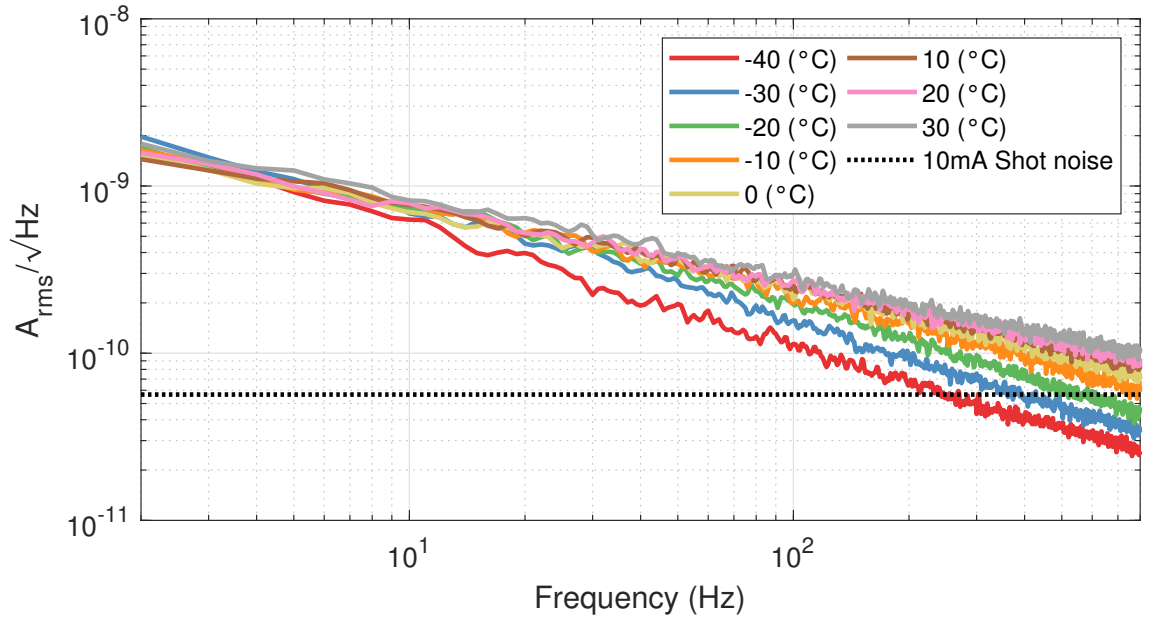


Figure A.13: IG24x500S4i 1.0V reverse bias

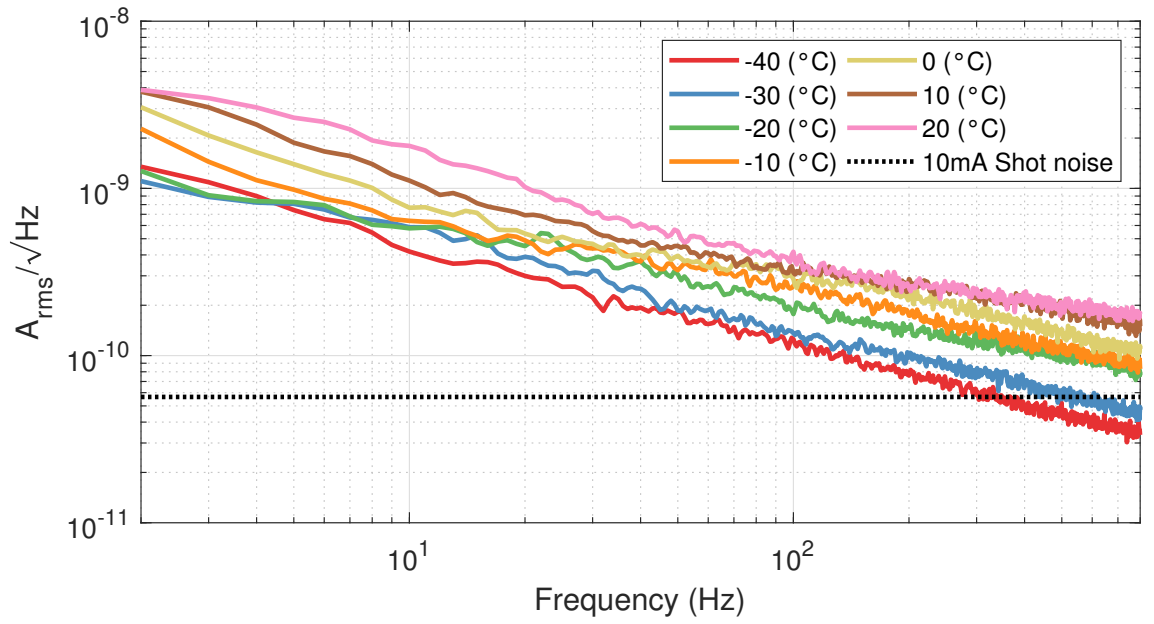


Figure A.14: IG26x500S4i 1.0V reverse bias

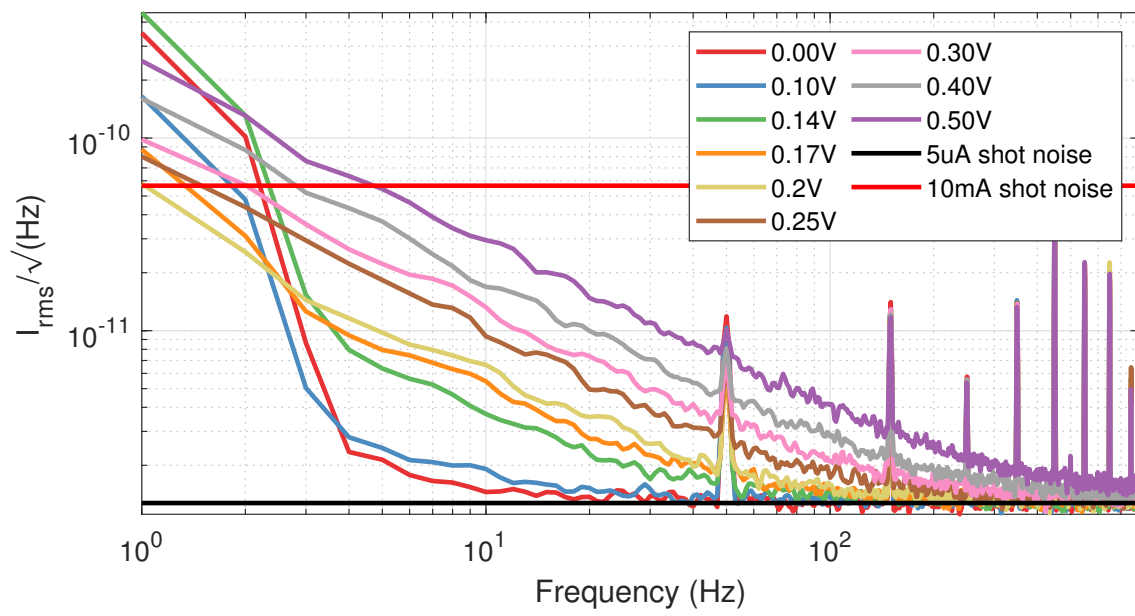


Figure A.15

Appendix B

Photodiode Circuits

B.1 Transimpedance Amplifier for Measuring the Dark Noise of Photodiodes

A circuit which allowed for the reverse bias to be set and for the easy exchange of photodiodes was constructed. To measure small amounts of current noise, a transimpedance amplifier with a large gain is needed. As the dark current for these diodes is of the order of 1mA, the gain of the transimpedance amplifier was chosen to be 1×10^6 so that signals of the order 1 V would be produced, and so a $1 \text{ M}\Omega$ resistor was used as in the feedback path.

Operation amplifiers have current noise at their inputs, voltage noise at the their outputs and the feedback resistor will generate Johnson-Nyquist noise (e.g. [105, Figure 8.58]). Designers of OP amps, generally, improve the OP amps output voltage noise at the expense of the input current noise or vice versa and so there is not one-size-fits-all op amp. In this application, as the current noise that is to be measured

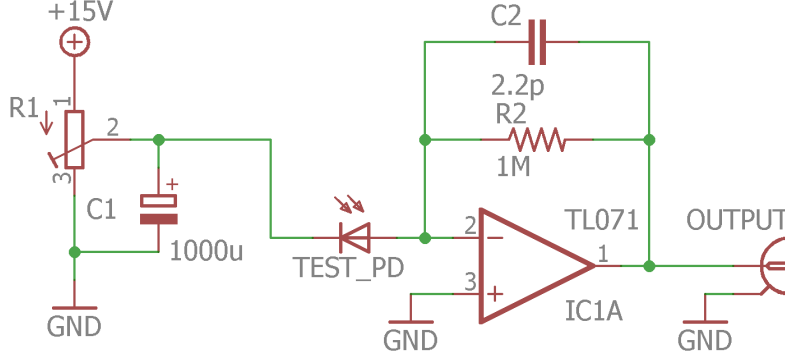


Figure B.1: Circuit used to measure the dark noise of photodiodes, represented by TEST_PD, under different reverse biases. The transimpedance amplifier has a $1\text{M}\Omega$ (R2) in parallel with a 2.2pF capacitor (C2) in the feedback path of the TL071 op amp (TL071P). The reverse bias was generated by a potentiometer (R1) connected to an LM7815 voltage regulator. This voltage was low passed using C1

is small and the feedback resistor is large, a TL071, a JFET style op amp with low input current noise, was selected. As the photodiode has capacitance and the op amp has finite bandwidth, a 2.2pF capacitor in parallel to keep the circuit stable.

Several methods of providing the reverse bias were tested. The first of these derived the bias from a LM7815 regulator in combination with a variable resistor configured a potential divider; this circuit is shown in Figure B.1. A similar circuit which was used in later experiments is shown in Figure B.2.

To verify that this was a sufficiently quiet bias voltage source that does not contaminate the measurements with noise, an ultra-quiet bias reference was used to cross check a measurement; this circuit is shown in Figure B.2. By filtering the voltage made a reference voltage IC (AD587JNZ), this circuit provided a bias with

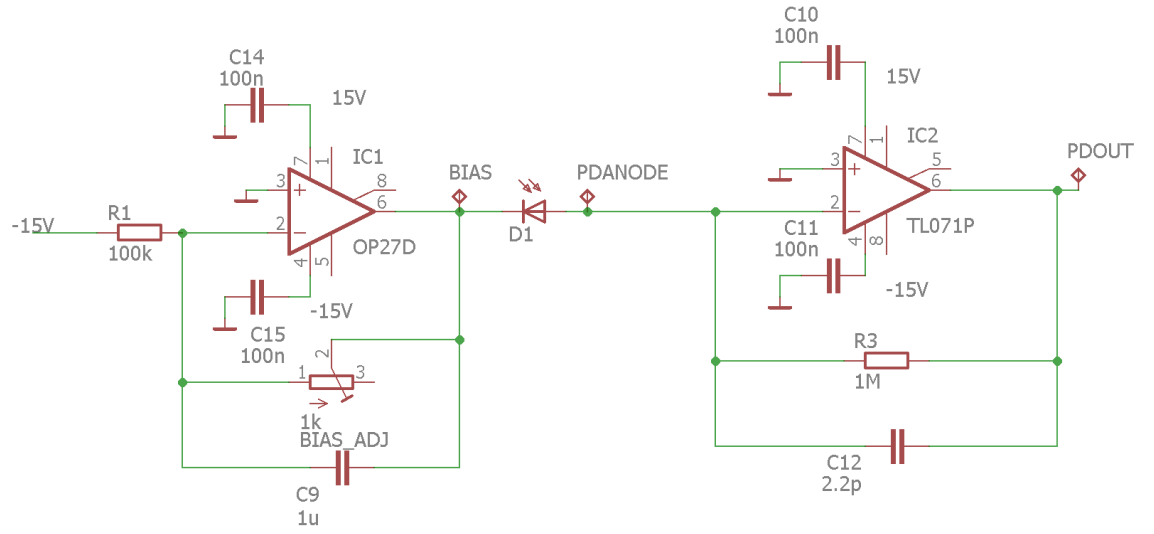


Figure B.2: Circuit used to measure the dark noise of photodiodes. The transimpedance amplifier used is identical to the one shown in Figure B.1. The bias voltage was generated using an OP27 op amp (OP27D) configured with so that the gain between the inverting input and the output of OP27D was > 1 .

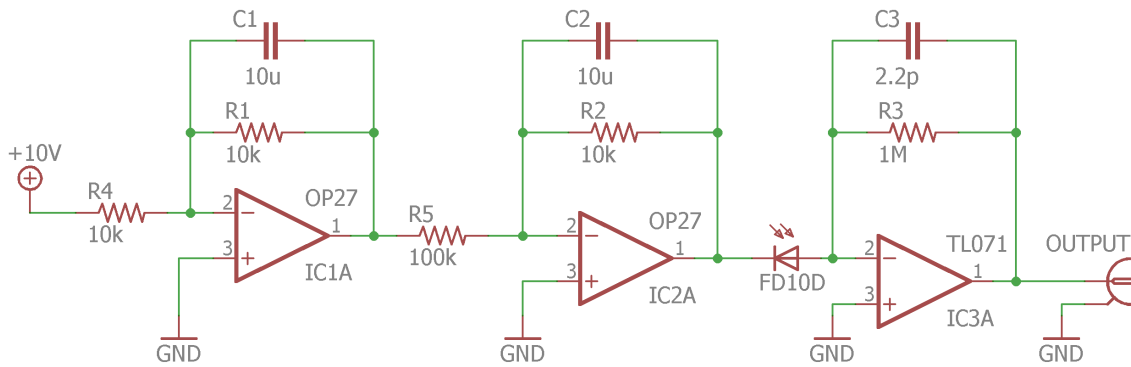


Figure B.3: Circuit used to check that bias voltage noise was not contaminating dark noise measurements. To reach $5\text{nV}/\sqrt{\text{Hz}}$ at 10 Hz, a 10V low noise reference IC (AD587JNZ) was used to generate the reverse bias. The output of this IC was filtered by two low pass filters (IC1A and IC2A). The transimpedance amplifier used is identical to the one shown in Figure B.1

$5\text{nV}/\sqrt{\text{Hz}}$ at 10 Hz. The bias of the FD10D (see Section ??) was set to 1 V in both circuits. Figure B.4 shows that the regulator-potentiometer bias voltage source is sufficiently quiet.

B.1.1 Transfer function

The transfer function of the transimpedance amplifiers shown in Figure ?? was measured under a variety of different physical constructions. This circuit's behaviour can be affected by stray capacitance. When constructed from strip board and with the photodiode anode and cathode cables very near each other, there was of the order 10 pF of stray capacitance i.e. it was as if the circuit had an extra 10 pF capacitor between the inverting input of the op amp and ground. By making this circuit on a prototyping board with less stray capacitance between the strips and

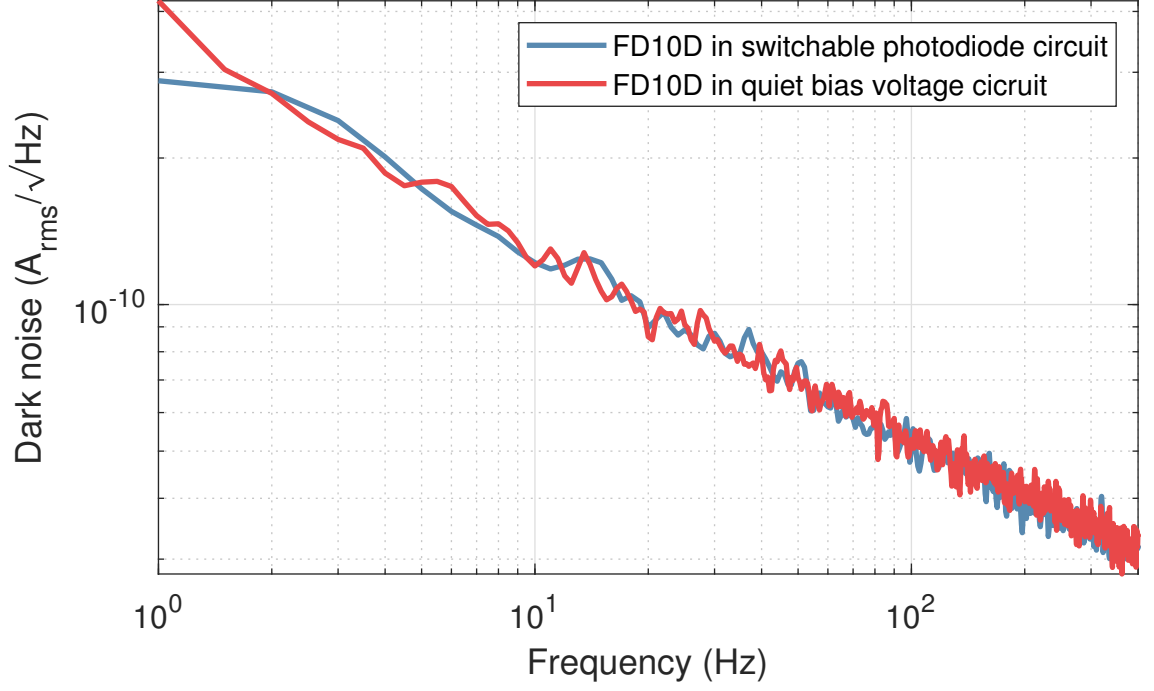


Figure B.4: The FD10D was biased with 1 V in the circuits shown in Figure B.1 and Figure B.2. The low noise fixed bias circuit and the circuit with a settable bias gave the same results, so we can be confident that the circuits shown in Figures B.1 and B.3 will produce measurements that are not contaminated by bias noise.

by keeping the photodiode anode and cathode wires from forming a capacitor, the stray capacitance was reduced. The measurement is in agreement with the LISO simulation of this circuit. This was cross checked using the method described in Section B.4.

B.1.2 Noise measurement

The dark noise of the circuits described in Section B.1 is shown in Figure B.6. The shot noise level of a typical photocurrent in a gravitational wave detector is several hundred times larger than the noise of this circuit. To measure noise of the transimpedance amplifier, instead of a photodiode, a 470 pF capacitor to was connected between the inverting input of the op amp and ground.

B.2 Transimpedance Amplifier for Making Shot Noise limited Measurements of 10 mA Photocurrents

A photodiode circuit which is shot noise limited over a up to several 100 kHz is shown in Figure B.7. A DC bias was created by low-passed bench-top lab power supply. An AC component could be added onto the bias via a summing OP-amp. The FD10D was connected to a transimpedance amplifier that whose electronic noise was below the shot noise level for ~ 10 mA currents. The noise of this circuit is shown in ... the transfer function of this circuit is shown in...

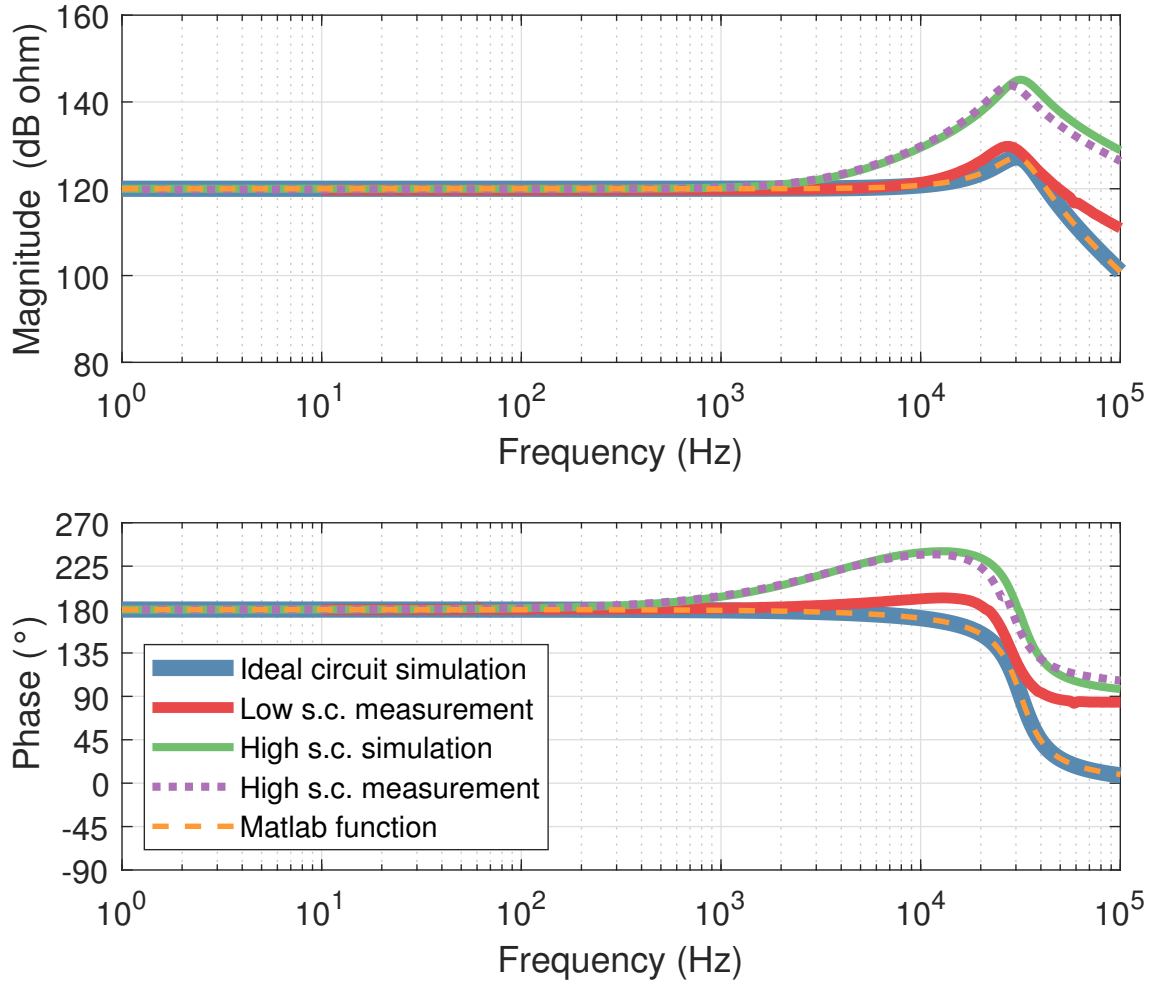


Figure B.5: The transfer function between a current into the inverting input of the transimpedance amplifiers designs shown in Figures B.1B.3 and B.2 was measured and simulated using LISO. A MATLAB function (see Section B.4) was used to check the measurement and the simulation. This circuit is sensitive to stray capacitance (s.c.) at the inverting input of the TL071. The stray capacitance was reduced by changing the layout of the circuit.

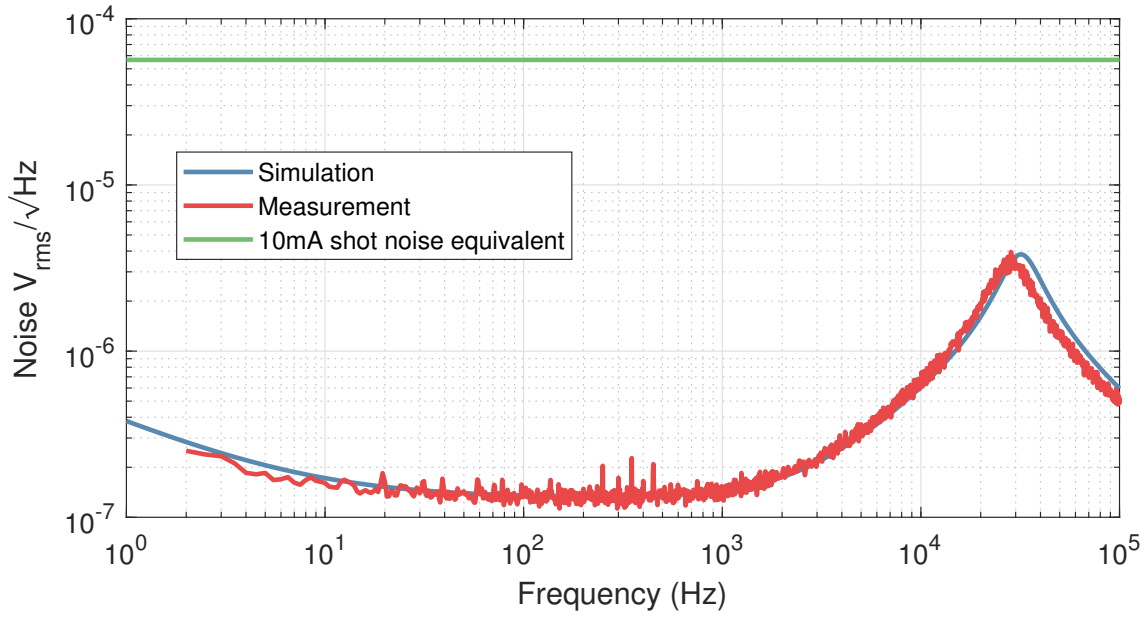


Figure B.6: At low frequencies, the noise of the circuit design shown in Figures B.1B.3 and B.2 is dominated by the low frequency voltage noise of the TL071. Above 10 Hz, the noise of the circuit is dominated by the johnson-Nyquist noise of the feedback resistor. At 30 kHz, the impedance of the feedback network causes the TL071's voltage noise to rise above the resistors noise.

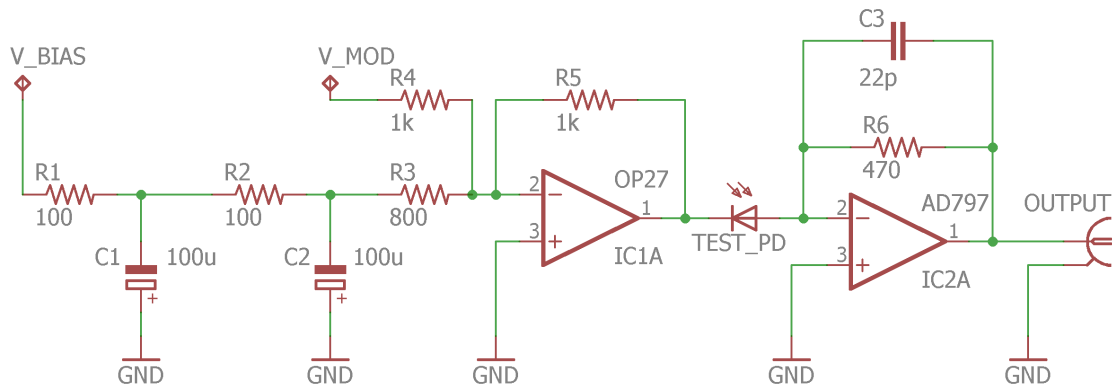


Figure B.7

B.3 Simulations with LISO

The behaviour of circuits was simulated using LISO. This software performs a nodal analysis to calculate the voltages at each point within a circuit. LISO can calculate the noise and transfer function of circuits that include op amps. LISO has a wide variety of op amps in its op amp library and this can be easily expanded by specifying the noise properties, open loop gain and gain-bandwidth frequency.

B.4 Transfer Function Calculation With MATLAB

The open loop transfer function of an op amp is

$$A(s) = \frac{A_0}{\frac{A_0}{2\pi f_{\text{GWB}}}s + 1}. \quad (\text{B.1})$$

The transfer function between the input current to the inverting input and the voltage at the output pin for an op amp made into a transimpedance amplifier is

$$H(s) = Z_{\text{FB}} \frac{A(s)}{1 + A(s)B(s)}, \quad (\text{B.2})$$

where $B(s)$ is the transfer function of the components between the output pin of the op amp and the inverting input of the op amp and Z_{FB} is the impedance seen by the current flowing from the output pin to the inverting input. Without a capacitor in parallel with the feedback resistor $B(s) = \frac{1}{R_{\text{FB}}C_{\text{PD}}s+1}$. This can be problematic as the transfer function of the circuit may have a phase of 180° due to the photodiode's capacitance and the gain bandwidth product of the op amp as it crosses the unity gain point which will cause the circuit to be unstable. To fix this problem, a capacitor

should be added in parallel to the feedback resistor, and so

$$B(s) = \frac{R_{\text{FB}}C_{\text{FB}}s + 1}{R_{\text{FB}}(C_{\text{FB}} + C_{\text{PD}})s + 1}. \quad (\text{B.3})$$

The impedance seen by the current is

$$Z_{\text{FB}} = \frac{R_{\text{FB}}}{R_{\text{FB}}(C_{\text{FB}} + C_{\text{PD}})s + 1}. \quad (\text{B.4})$$

$H(s)$ can be calculated with the following MATLAB code:

```
function [mag,phase] = photodiodeTransimpedance(cf, rf, cpd, gbw, A, w)
% cf is the capacitor in the feedback loop used to keep the circuit
% stable, and its units are Farads. rf is the feedback resistor
% measured in Ohms. cpd is the photodiodes capacitance (Farads).
    gbw is the
% gain bandwidth of the OP amp, and is measured in Hz. A is the
    Open loop
% gain of the OP amp. w is the angular frequency range that
% the phase and magnitude will be calculated over.
A_s = tf(A, [A/(2*pi*gbw), 1]);
B_s = tf([cf*rf, 1], [rf*(cf+cpd), 1]);
closedLoopAnalytic = -tf(rf, [rf*(cf+cpd), 1])*A_s/(1+A_s*B_s);
[mag, phase] = bode(closedLoopAnalytic, w);
mag = squeeze(mag);
phase = squeeze(phase);
end
```

Bibliography

- [1] LIGO Virgo Collaboration. Observation of gravitational waves from a binary black hole merger. *Phys. Rev. Lett.*, 116:061102, Feb 2016.
- [2] The LIGO Scientific Collaboration. Advanced LIGO. *Classical and Quantum Gravity*, 32(7):074001, mar 2015.
- [3] LIGO Virgo Collaboration. Gw170814: A three-detector observation of gravitational waves from a binary black hole coalescence. *Phys. Rev. Lett.*, 119:141101, Oct 2017.
- [4] The LIGO Virgo Collaboration. GW190412: Observation of a binary-black-hole coalescence with asymmetric masses. *Physical Review D*, 102(4), aug 2020.
- [5] The LIGO Virgo Collaboration. GW190521: A binary black hole merger with a total mass of $150\text{ m}_{\odot}$. *Physical Review Letters*, 125(10), sep 2020.
- [6] LIGO Virgo Collaboration. Gw170817: Observation of gravitational waves from a binary neutron star inspiral. *Phys. Rev. Lett.*, 119:161101, Oct 2017.

- [7] The LIGO Virgo Collaboration. GW190425: Observation of a compact binary coalescence with total mass $\sim 3.4 \text{ m } \odot$. *The Astrophysical Journal*, 892(1):L3, mar 2020.
- [8] The LVK collaboration. GW190814: Gravitational waves from the coalescence of a 23 solar mass black hole with a 2.6 solar mass compact object. *The Astrophysical Journal*, 896(2):L44, jun 2020.
- [9] The LIGO Virgo Collaboration. Gwtc-2: Compact binary coalescences observed by ligo and virgo during the first half of the third observing run.
- [10] The LIGO Virgo Collaboration. The basic physics of the binary black hole merger GW150914. *Annalen der Physik*, 529(1-2):1600209, oct 2016.
- [11] The LIGO Scientific Collaboration and the Virgo Collaboration. Tests of general relativity with binary black holes from the second ligo-virgo gravitational-wave transient catalog.
- [12] Charles W. Misner. *Gravitation*. Princeton University Press, oct 2017.
- [13] Bernard Schutz. *A First Course in General Relativity; 2nd edition*. Cambridge University Press, 2009.
- [14] LVK collaboration. Narrow-band search for gravitational waves from known pulsars using the second LIGO observing run. *Physical Review D*, 99(12), jun 2019.
- [15] The LIGO Virgo Collaboration. Searches for continuous gravitational waves from 15 supernova remnants and fomalhaut b with advanced LIGO. *The Astrophysical Journal*, 875(2):122, apr 2019.

- [16] The LIGO Virgo Collaboration. Search for transient gravitational-wave signals associated with magnetar bursts during advanced LIGO’s second observing run. *The Astrophysical Journal*, 874(2):163, apr 2019.
- [17] B. S. Sathyaprakash and Bernard F. Schutz. Physics, astrophysics and cosmology with gravitational waves. *Living Reviews in Relativity*, 12(1), mar 2009.
- [18] Ernazar Abdikamalov, Giulia Pagliaroli, and David Radice. Gravitational waves from core-collapse supernovae.
- [19] Teng Zhang, Jiří Smetana, Yikang Chen, Joe Bentley, William E. East, Denis Martynov, Haixing Miao, and Huan Yang. Towards observing the neutron star collapse with gravitational wave detectors.
- [20] The LIGO Virgo Collaboration. Search for the isotropic stochastic background using data from advanced LIGO’s second observing run. *Physical Review D*, 100(6), sep 2019.
- [21] Juan García-Bellido. Massive primordial black holes as dark matter and their detection with gravitational waves. *Journal of Physics: Conference Series*, 840:012032, may 2017.
- [22] Simeon Bird, Ilias Cholis, Julian B. Muñoz, Yacine Ali-Haïmoud, Marc Kamionkowski, Ely D. Kovetz, Alvise Raccanelli, and Adam G. Riess. Did LIGO detect dark matter? *Physical Review Letters*, 116(20), may 2016.
- [23] T. Regimbau, M. Evans, N. Christensen, E. Katsavounidis, B. Sathyaprakash, and S. Vitale. Digging deeper: Observing primordial gravitational waves be-

- low the binary-black-hole-produced stochastic background. *Physical Review Letters*, 118(15), apr 2017.
- [24] Yanou Cui, Marek Lewicki, David E. Morrissey, and James D. Wells. Probing the pre-BBN universe with gravitational waves from cosmic strings. *Journal of High Energy Physics*, 2019(1), jan 2019.
- [25] The LIGO Scientific Collaboration, the Virgo Collaboration, and the KAGRA Collaboration. Prospects for observing and localizing gravitational-wave transients with advanced ligo, advanced virgo and kagra, version 10. 2020.
- [26] F Acernese et al. Advanced virgo: a second-generation interferometric gravitational wave detector. *Classical and Quantum Gravity*, 32(2):024001, dec 2014.
- [27] Yoichi Aso, Yuta Michimura, Kentaro Somiya, Masaki Ando, Osamu Miyakawa, Takanori Sekiguchi, Daisuke Tatsumi, and Hiroaki Yamamoto. Interferometer design of the KAGRA gravitational wave detector. *Physical Review D*, 88(4), aug 2013.
- [28] S M Aston, M A Barton, A S Bell, N Beveridge, B Bland, A J Brummitt, G Cagnoli, C A Cantley, L Carbone, A V Cumming, L Cunningham, R M Cutler, R J S Greenhalgh, G D Hammond, K Haughian, T M Hayler, A Heptonstall, J Heefner, D Hoyland, J Hough, R Jones, J S Kissel, R Kumar, N A Lockerbie, D Lodhia, I W Martin, P G Murray, J O’Dell, M V Plissi, S Reid, J Romie, N A Robertson, S Rowan, B Shapiro, C C Speake, K A Strain, K V Tokmakov, C Torrie, A A van Veggel, A Vecchio, and I Wilmot. Update

on quadruple suspension design for advanced LIGO. *Classical and Quantum Gravity*, 29(23):235004, oct 2012.

- [29] Gregory M Harry, Matthew R Abernathy, Andres E Becerra-Toledo, Helena Armandula, Eric Black, Kate Dooley, Matt Eichenfield, Chinyere Nwabugwu, Akira Villar, D R M Crooks, Gianpietro Cagnoli, Jim Hough, Colin R How, Ian MacLaren, Peter Murray, Stuart Reid, Sheila Rowan, Peter H Sneddon, Martin M Fejer, Roger Route, Steven D Penn, Patrick Ganau, Jean-Marie Mackowski, Christophe Michel, Laurent Pinard, and Alban Remillieux. Titania-doped tantala/silica coatings for gravitational-wave detection. *Classical and Quantum Gravity*, 24(2):405–415, dec 2006.
- [30] S Hild, H Grote, J Degallaix, S Chelkowski, K Danzmann, A Freise, M Hewitson, J Hough, H Lück, M Prijatelj, K A Strain, J R Smith, and B Willke. DC-readout of a signal-recycled gravitational wave detector. *Classical and Quantum Gravity*, 26(5):055012, feb 2009.
- [31] Tobin T Fricke, Nicolás D Smith-Lefebvre, Richard Abbott, Rana Adhikari, Katherine L Dooley, Matthew Evans, Peter Fritschel, Valery V Frolov, Keita Kawabe, Jeffrey S Kissel, Bram J J Slagmolen, and Sam J Waldman. DC readout experiment in enhanced LIGO. *Classical and Quantum Gravity*, 29(6):065005, feb 2012.
- [32] J. Mizuno. *Comparison of optical configurations for laser-interferometric gravitational-wave detectors*. PhD thesis, AEI-Hannover, MPI for Gravitational Physics, Max Planck Society, 1995.

- [33] Gerhard Heinzel. Advanced optical techniques for laser-interferometric gravitational-wave detectors, 1999.
- [34] B.J. Meers. The frequency response of interferometric gravitational wave detectors. *Physics Letters A*, 142(8-9):465–470, dec 1989.
- [35] Charlotte Bond, Daniel Brown, Andreas Freise, and Kenneth A. Strain. Interferometer techniques for gravitational-wave detection. *Living Reviews in Relativity*, 19(1), dec 2016.
- [36] Vladimir B. Braginsky. Energetic quantum limit in large-scale interferometers. In *AIP Conference Proceedings*. AIP, 2000.
- [37] Carlton M. Caves. Quantum-mechanical noise in an interferometer. *Physical Review D*, 23(8):1693–1708, apr 1981.
- [38] Stefan L. Danilishin and Farid Ya. Khalili. Quantum measurement theory in gravitational-wave detectors. *Living Reviews in Relativity*, 15(1), apr 2012.
- [39]
- [40] Brian J. Meers. Recycling in laser-interferometric gravitational-wave detectors. *Physical Review D*, 38(8):2317–2326, oct 1988.
- [41] K. A. Strain and B. J. Meers. Experimental demonstration of dual recycling for interferometric gravitational-wave detectors. *Physical Review Letters*, 66(11):1391–1394, mar 1991.
- [42] D. Schnier, J. Mizuno, G. Heinzel, H. Lück, A. Rüdiger, R. Schilling, M. Schrempel, W. Winkler, and K. Danzmann. Power recycling in the garch-

- ing 30 m prototype interferometer for gravitational-wave detection. *Physics Letters A*, 225(4-6):210–216, feb 1997.
- [43] R. W. P. Drever, J. L. Hall, F. V. Kowalski, J. Hough, G. M. Ford, A. J. Munley, and H. Ward. Laser phase and frequency stabilization using an optical resonator. *Applied Physics B*, 31(2):97–105, Jun 1983.
- [44] Muzammil A. Arain and Guido Mueller. Design of the advanced LIGO recycling cavities. *Optics Express*, 16(14):10018, jun 2008.
- [45] G. Heinzel, J. Mizuno, R. Schilling, W. Winkler, A. Rüdiger, and K. Danzmann. An experimental demonstration of resonant sideband extraction for laser-interferometric gravitational wave detectors. *Physics Letters A*, 217(6):305–314, jul 1996.
- [46] J. Mizuno, K.A. Strain, P.G. Nelson, J.M. Chen, R. Schilling, A. Rüdiger, W. Winkler, and K. Danzmann. Resonant sideband extraction: a new configuration for interferometric gravitational wave detectors. *Physics Letters A*, 175(5):273–276, apr 1993.
- [47] Dhruva Ganapathy, Lee McCuller, Jameson Graef Rollins, Evan D. Hall, Lisa Barsotti, and Matthew Evans. Tuning advanced ligo to kilohertz signals from neutron-star collisions.
- [48] Peter Fritschel Jeff Lewis Sam Waldman Koji Arai, Sam Barnum. LIGO-T1000276-v5.

- [49] M Prijatelj, J Degallaix, H Grote, J Leong, C Affeldt, S Hild, H Lück, J Slutsky, H Wittel, K Strain, and K Danzmann. The output mode cleaner of GEO 600. *Classical and Quantum Gravity*, 29(5):055009, feb 2012.
- [50] H. J. Kimble, Yuri Levin, Andrey B. Matsko, Kip S. Thorne, and Sergey P. Vyatchanin. Conversion of conventional gravitational-wave interferometers into quantum nondemolition interferometers by modifying their input and/or output optics. *Physical Review D*, 65(2), dec 2001.
- [51] V. B. Braginsky and F. Ya. Khalili. Quantum nondemolition measurements: the route from toys to tools. *Reviews of Modern Physics*, 68(1):1–11, jan 1996.
- [52] Alessandra Buonanno and Yanbei Chen. Quantum noise in second generation, signal-recycled laser interferometric gravitational-wave detectors. *Physical Review D*, 64(4), jul 2001.
- [53] R Flaminio, J Franc, C Michel, N Morgado, L Pinard, and B Sassolas. A study of coating mechanical and optical losses in view of reducing mirror thermal noise in gravitational wave detectors. *Classical and Quantum Gravity*, 27(8):084030, apr 2010.
- [54] Rainer Weiss. Quarterly progress report: Electromagnetically coupled broadband gravitational antenna. DCC LIGO-P720002-v1, April 1972.
- [55] A. Buikema, C. Cahillane, G.L. Mansell, C.D. Blair, R. Abbott, C. Adams, R.X. Adhikari, A. Ananyeva, S. Appert, K. Arai, J.S. Areeda, Y. Asali, S.M. Aston, C. Austin, A.M. Baer, M. Ball, S.W. Ballmer, S. Banagiri, D. Barker, L. Barsotti, J. Bartlett, B.K. Berger, J. Betzwieser, D. Bhattacharjee, G. Billingsley, S. Biscans, R. M. Blair, N. Bode, P. Booker, R. Bork,

A. Bramley, A.F. Brooks, D.D. Brown, K.C. Cannon, X. Chen, A.A. Ciobanu, F. Clara, S.J. Cooper, K.R. Corley, S.T. Countryman, P.B. Co-
 vas, D.C. Coyne, L.E.H. Datrier, D. Davis, C. Di Fronzo, K.L. Dooley, J.C.
 Driggers, P. Dupej, S.E. Dwyer, A. Effler, T. Etzel, M. Evans, T.M. Evans,
 J. Feicht, A. Fernandez-Galiana, P. Fritschel, V.V. Frolov, P. Fulda, M. Fyffe,
 J. A. Giaime, K.D. Giardina, P. Godwin, E. Goetz, S. Gras, C. Gray, R. Gray,
 A.C. Green, E.K. Gustafson, R. Gustafson, J. Hanks, J. Hanson, T. Hard-
 wick, R.K. Hasskew, M.C. Heintze, A.F. Helmling-Cornell, N.A. Holland, J.
 D. Jones, S. Kandhasamy, S. Karki, M. Kasprzack, K. Kawabe, N. Kijbun-
 choo, P.J. King, J.S. Kissel, Rahul Kumar, M. Landry, B.B. Lane, B. Lantz,
 M. Laxen, Y.K. Lecoecuche, J. Leviton, J. Liu, M. Lormand, A.P. Lundgren,
 R. Macas, M. MacInnis, D.M. Macleod, S. Márka, Z. Márka, D.V. Martynov,
 K. Mason, T.J. Massinger, F. Matichard, N. Mavalvala, R. McCarthy, D.E.
 McClelland, S. McCormick, L. McCuller, J. McIver, T. McRae, G. Mendell,
 K. Merfeld, E.L. Merilh, F. Meylahn, T. Mistry, R. Mittleman, G. Moreno,
 C.M. Mow-Lowry, S. Mozzon, A. Mullavey, T.J.N. Nelson, P. Nguyen, L.
 K. Nuttall, J. Oberling, Richard J. Oram, B. O'Reilly, C. Osthelder, D.J.
 Ottaway, H. Overmier, J.R. Palamos, W. Parker, E. Payne, A. Pele, R. Pen-
 horwood, C.J. Perez, M. Pirello, H. Radkins, K.E. Ramirez, J.W. Richardson,
 K. Riles, N.A. Robertson, J.G. Rollins, C.L. Romel, J.H. Romie, M.P. Ross,
 K. Ryan, T. Sadecki, E.J. Sanchez, L.E. Sanchez, T.R. Saravanan, R.L.
 Savage, D. Schaetzel, R. Schnabel, R.M.S. Schofield, E. Schwartz, D. Sellers,
 T. Shaffer, D. Sigg, B.J.J. Slagmolen, J.R. Smith, S. Soni, B. Sorazu, A.P.
 Spencer, K.A. Strain, L. Sun, M.J. Szczepańczyk, M. Thomas, P. Thomas,
 K.A. Thorne, K. Toland, C.I. Torrie, G. Traylor, M. Tse, A.L. Urban, G. Va-

- jente, G. Valdes, D. C. Vander-Hyde, P. J. Veitch, K. Venkateswara, G. Venugopalan, A. D. Viets, T. Vo, C. Vorvick, M. Wade, R. L. Ward, J. Warner, B. Weaver, R. Weiss, C. Whittle, B. Willke, C. C. Wipf, L. Xiao, H. Yamamoto, Hang Yu, Haocun Yu, L. Zhang, M. E. Zucker, and J. Zweizig. Sensitivity and performance of the advanced LIGO detectors in the third observing run. *Physical Review D*, 102(6), sep 2020.
- [56] Carlton M. Caves and Bonny L. Schumaker. New formalism for two-photon quantum optics. i. quadrature phases and squeezed states. *Physical Review A*, 31(5):3068–3092, may 1985.
- [57] Bonny L. Schumaker and Carlton M. Caves. New formalism for two-photon quantum optics. II. mathematical foundation and compact notation. *Physical Review A*, 31(5):3093–3111, may 1985.
- [58] M. Tse, Haocun Yu, N. Kijbunchoo, A. Fernandez-Galiana, P. Dupej, L. Barsotti, C. D. Blair, D. D. Brown, S. E. Dwyer, A. Effler, M. Evans, P. Fritschel, V. V. Frolov, A. C. Green, G. L. Mansell, F. Matichard, N. Mavalvala, D. E. McClelland, L. McCuller, T. McRae, J. Miller, A. Mullavey, E. Oelker, I. Y. Phinney, D. Sigg, B. J. J. Slagmolen, T. Vo, R. L. Ward, C. Whittle, R. Abbott, C. Adams, R. X. Adhikari, A. Ananyeva, S. Appert, K. Arai, J. S. Areeda, Y. Asali, S. M. Aston, C. Austin, A. M. Baer, M. Ball, S. W. Ballmer, S. Bagnagiri, D. Barker, J. Bartlett, B. K. Berger, J. Betzwieser, D. Bhattacharjee, G. Billingsley, S. Biscans, R. M. Blair, N. Bode, P. Booker, R. Bork, A. Bramley, A. F. Brooks, A. Buikema, C. Cahillane, K. C. Cannon, X. Chen, A. A. Ciobanu, F. Clara, S. J. Cooper, K. R. Corley, S. T. Countryman, P. B. Covas, D. C. Coyne, L. E. H. Datrier, D. Davis, C. Di Fronzo, J. C. Driggers,

T. Etzel, T. M. Evans, J. Feicht, P. Fulda, M. Fyffe, J. A. Giaime, K. D. Giardina, P. Godwin, E. Goetz, S. Gras, C. Gray, R. Gray, Anchal Gupta, E. K. Gustafson, R. Gustafson, J. Hanks, J. Hanson, T. Hardwick, R. K. Hasskew, M. C. Heintze, A. F. Helmling-Cornell, N. A. Holland, J. D. Jones, S. Kandhasamy, S. Karki, M. Kasprzack, K. Kawabe, P. J. King, J. S. Kissel, Rahul Kumar, M. Landry, B. B. Lane, B. Lantz, M. Laxen, Y. K. Lecoecue, J. Leviton, J. Liu, M. Lormand, A. P. Lundgren, R. Macas, M. MacInnis, D. M. Macleod, S. Márka, Z. Márka, D. V. Martynov, K. Mason, T. J. Massinger, R. McCarthy, S. McCormick, J. McIver, G. Mendell, K. Merfeld, E. L. Merilh, F. Meylahn, T. Mistry, R. Mittleman, G. Moreno, C. M. Mow-Lowry, S. Mozzon, T. J. N. Nelson, P. Nguyen, L. K. Nuttall, J. Oberling, R. J. Oram, B. O'Reilly, C. Osthelder, D. J. Ottaway, H. Overmier, J. R. Palamos, W. Parker, E. Payne, A. Pele, C. J. Perez, M. Pirello, H. Radkins, K. E. Ramirez, J. W. Richardson, K. Riles, N. A. Robertson, J. G. Rollins, C. L. Romel, J. H. Romie, M. P. Ross, K. Ryan, T. Sadecki, E. J. Sanchez, L. E. Sanchez, T. R. Saravanan, R. L. Savage, D. Schaetzl, R. Schnabel, R. M. S. Schofield, E. Schwartz, D. Sellers, T. J. Shaffer, J. R. Smith, S. Soni, B. Sorazu, A. P. Spencer, K. A. Strain, L. Sun, M. J. Szczepańczyk, M. Thomas, P. Thomas, K. A. Thorne, K. Toland, C. I. Torrie, G. Traylor, A. L. Urban, G. Vajente, G. Valdes, D. C. VanderHyde, P. J. Veitch, K. Venkateswara, G. Venugopalan, A. D. Viets, C. Vorvick, M. Wade, J. Warner, B. Weaver, R. Weiss, B. Willke, C. C. Wipf, L. Xiao, H. Yamamoto, M. J. Yap, Hang Yu, L. Zhang, M. E. Zucker, and J. Zweizig. Quantum-enhanced advanced LIGO detectors in the era of gravitational-wave astronomy. *Physical Review Letters*, 123(23), dec 2019.

- [59] James Lough, Emil Schreiber, Fabio Bergamin, Hartmut Grote, Moritz Mehmet, Henning Vahlbruch, Christoph Affeldt, Marc Brinkmann, Aparna Bisht, Volker Kringel, Harald Lück, Nikhil Mukund, Séverin Nadji, Borja Sorazu, Kenneth Strain, Michael Weinert, and Karsten Danzmann. First demonstration of 6 db quantum noise reduction in a kilometer scale gravitational wave observatory.
- [60] D. F. Walls. Squeezed states of light. *Nature*, 306(5939):141–146, nov 1983.
- [61] Haocun Yu, , L. McCuller, M. Tse, N. Kijbunchoo, L. Barsotti, and N. Mavalvala. Quantum correlations between light and the kilogram-mass mirrors of LIGO. *Nature*, 583(7814):43–47, jul 2020.
- [62] Kentaro Komori, Dhruva Ganapathy, Chris Whittle, Lee McCuller, Lisa Barsotti, Nergis Mavalvala, and Matthew Evans. Demonstration of an amplitude filter cavity at gravitational-wave frequencies. *Physical Review D*, 102(10), nov 2020.
- [63] Joe Bentley, Philip Jones, Denis Martynov, Andreas Freise, and Haixing Miao. Converting the signal-recycling cavity into an unstable optomechanical filter to enhance the detection bandwidth of gravitational-wave detectors. *Physical Review D*, 99(10), may 2019.
- [64] Haixing Miao, Yiqiu Ma, Chunnong Zhao, and Yanbei Chen. Enhancing the bandwidth of gravitational-wave detectors with unstable optomechanical filters. *Physical Review Letters*, 115(21), nov 2015.

- [65] Xiang Li, Maxim Goryachev, Yiqiu Ma, Michael E. Tobar, Chunrong Zhao, Rana X Adhikari, , and Yanbei Chen. Broadband sensitivity improvement via coherent quantum feedback with pt symmetry. *PRL*, To be published.
- [66] Vladimir B. Braginsky, Yuri I. Vorontsov, and Kip S. Thorne. Quantum non-demolition measurements. *Science*, 209(4456):547–557, 1980.
- [67] Patricia Purdue and Yanbei Chen. Practical speed meter designs for quantum nondemolition gravitational-wave interferometers. *Physical Review D*, 66(12), dec 2002.
- [68] T. Zhang, E. Knyazev, S. Steinlechner, F. Y. Khalili, B. W. Barr, A. S. Bell, P. Dupej, J. Briggs, C. Gräf, J. Callaghan, J. S. Hennig, E. A. Houston, S. H. Huttner, S. S. Leavey, D. Pascucci, B. Sorazu, A. Spencer, J. Wright, K. A. Strain, S. Hild, and S. L. Danilishin. Quantum noise cancellation in asymmetric speed metres with balanced homodyne readout. *New journal of physics*, 20(10):103040, 2018.
- [69] S H Huttner, S L Danilishin, S Hild, and K A Strain. Comparison of different sloshing speedmeters. *Classical and Quantum Gravity*, 37(8):085022, mar 2020.
- [70] A. Brookes, L. Barsotti, G. Billingsley, K.A. Strain, and M. Zucker. A+ o5-ham6 design considerations and requirements. LIGO DCC, April 2020. E1900385-v1.
- [71] L. McCuller, C. Whittle, D. Ganapathy, K. Komori, M. Tse, A. Fernandez-Galiana, L. Barsotti, P. Fritschel, M. MacInnis, F. Matichard, K. Mason,

- N. Mavalvala, R. Mittleman, Haocun Yu, M.E. Zucker, and M. Evans. Frequency-dependent squeezing for advanced LIGO. *Physical Review Letters*, 124(17), apr 2020.
- [72] P. Fritschel and E. Gustafson. Advanced LIGO Optical Layout. LIGO Document D0902838-v5 <https://dcc.ligo.org/LIGO-D0902838>.
- [73] L. Barsotti, Gras. S, M. Evans, and P. Fritschel. Updated Advanced LIGO sensitivity design curve . LIGO DCC, April 2018. LIGO Document T1800044-v5 <https://dcc.ligo.org/LIGO-T1800044/public>.
- [74] L. Barsotti, L. McCuller, M. Evans, and P. Fritschel. The A+ design curve. LIGO DCC, April 2018. LIGO Document T1800042-v5 <https://dcc.ligo.org/LIGO-T1800042/public>.
- [75] M Punturo, M Abernathy, F Acernese, B Allen, N Andersson, K Arun, F Barone, B Barr, M Barsuglia, M Beker, N Beveridge, S Birindelli, S Bose, L Bosi, S Braccini, C Bradaschia, T Bulik, E Calloni, G Cella, E Chassande Mottin, S Chelkowski, A Chincarini, J Clark, E Coccia, C Colacino, J Colas, A Cumming, L Cunningham, E Cuoco, S Danilishin, K Danzmann, G De Luca, R De Salvo, T Dent, R De Rosa, L Di Fiore, A Di Virgilio, M Doets, V Fafone, P Falferi, R Flaminio, J Franc, F Frasconi, A Freise, P Fulda, J Gair, G Gemme, A Gennai, A Giazotto, K Glampedakis, M Granata, H Grote, G Guidi, G Hammond, M Hannam, J Harms, D Heinert, M Hendry, I Heng, E Hennes, S Hild, J Hough, S Husa, S Huttner, G Jones, F Khalili, K Kokeyama, K Kokkotas, B Krishnan, M Lorenzini, H Lück, E Majorana, I Mandel, V Mandic, I Martin, C Michel, Y Minenkov, N Morgado,

S Mosca, B Mours, H Müller–Ebhardt, P Murray, R Nawrodt, J Nelson, R Oshaughnessy, C D Ott, C Palomba, A Paoli, G Parguez, A Pasqualetti, R Passaquieti, D Passuello, L Pinard, R Poggiani, P Popolizio, M Prato, P Puppo, D Rabeling, P Rapagnani, J Read, T Regimbau, H Rehbein, S Reid, L Rezzolla, F Ricci, F Richard, A Rocchi, S Rowan, A Rüdiger, B Sassolas, B Sathyaprakash, R Schnabel, C Schwarz, P Seidel, A Sintes, K Somiya, F Speirits, K Strain, S Strigin, P Sutton, S Tarabrin, A Thüring, J van den Brand, C van Leewen, M van Veggel, C van den Broeck, A Vecchio, J Veitch, F Vetrano, A Vicere, S Vyatchanin, B Willke, G Woan, P Wolfango, and K Yamamoto. The einstein telescope: a third-generation gravitational wave observatory. *Classical and Quantum Gravity*, 27(19):194002, sep 2010.

- [76] R X Adhikari, K Arai, A F Brooks, C Wipf, O Aguiar, P Altin, B Barr, L Barsotti, R Bassiri, A Bell, G Billingsley, R Birney, D Blair, E Bonilla, J Briggs, D D Brown, R Byer, H Cao, M Constancio, S Cooper, T Corbitt, D Coyne, A Cumming, E Daw, R deRosa, G Eddolls, J Eichholz, M Evans, M Fejer, E C Ferreira, A Freise, V V Frolov, S Gras, A Green, H Grote, E Gustafson, E D Hall, G Hammond, J Harms, G Harry, K Haughian, D Heinert, M Heintze, F Hellman, J Hennig, M Hennig, S Hild, J Hough, W Johnson, B Kamai, D Kapasi, K Komori, D Koptsov, M Korobko, W Z Korth, K Kuns, B Lantz, S Leavey, F Magana-Sandoval, G Mansell, A Markosyan, A Markowitz, I Martin, R Martin, D Martynov, D E McClelland, G McGhee, T McRae, J Mills, V Mitrofanov, M Molina-Ruiz, C Mow-Lowry, J Munch, P Murray, S Ng, M A Okada, D J Ottaway, L Prokhorov, V Quetschke, S Reid, D Reitze, J Richardson, R Robie, I Romero-Shaw, R Route, S Rowan, R Schnabel, M Schneewind,

- F Seifert, D Shaddock, B Shapiro, D Shoemaker, A S Silva, B Slagmolen, J Smith, N Smith, J Steinlechner, K Strain, D Taira, S Tait, D Tanner, Z Tornasi, C Torrie, M Van Veggel, J Vanheijningen, P Veitch, A Wade, G Wallace, R Ward, R Weiss, P Wessels, B Willke, H Yamamoto, M J Yap, and C Zhao. A cryogenic silicon interferometer for gravitational-wave detection. *Classical and Quantum Gravity*, 37(16):165003, jul 2020.
- [77] David Reitze, Rana X Adhikari, Stefan Ballmer, Barry Barish, Lisa Barsotti, GariLynn Billingsley, Duncan A. Brown, Yanbei Chen, Dennis Coyne, Robert Eisenstein, Matthew Evans, Peter Fritschel, Evan D. Hall, Albert Lazzarini, Geoffrey Lovelace, Jocelyn Read, B. S. Sathyaprakash, David Shoemaker, Joshua Smith, Calum Torrie, Salvatore Vitale, Rainer Weiss, Christopher Wipf, and Michael Zucker. Cosmic explorer: The u.s. contribution to gravitational-wave astronomy beyond ligo.
- [78] Kieran Craig, Jessica Steinlechner, Peter G. Murray, Angus S. Bell, Ross Birney, Karen Haughian, Jim Hough, Ian MacLaren, Steve Penn, Stuart Reid, Raymond Robie, Sheila Rowan, and Iain W. Martin. Mirror coating solution for the cryogenic einstein telescope. *Physical Review Letters*, 122(23), jun 2019.
- [79] B. Willke, P. King, R. Savage, and P. Fritschel. Pre-stabilized laser design requirements. DCC LIGO-T050036-v4, July 2011.
- [80] E. A. Shcherbakov, V. V. Fomin, A. A. Abramov, A. A. Ferin, D. V. Mochalov, and V. P. Gapontsev. Industrial grade 100 kW power CW fiber laser. In *Advanced Solid-State Lasers Congress*. OSA, 2013.

- [81] Junsu Lee, Kwang Hyun Lee, Hwanseong Jeong, Minkyu Park, Ji Hoon Seung, and Jung Hwan Lee. 205 kW all-fiber high-beam-quality fiber amplifier with stimulated brillouin scattering suppression incorporating a narrow-linewidth fiber-bragg-grating-stabilized laser diode seed source. *Applied Optics*, 58(23):6251, aug 2019.
- [82] Yusheng Huang, Ping Yan, Zehui Wang, Jiading Tian, Dan Li, Qirong Xiao, and Mali Gong. 219 kW narrow linewidth FBG-based MOPA configuration fiber laser. *Optics Express*, 27(3):3136, jan 2019.
- [83] Pengfei Ma, Rumao Tao, Rongtao Su, Xiaolin Wang, Pu Zhou, and Zejin Liu. 189 kW all-fiberized and polarization-maintained amplifiers with narrow linewidth and near-diffraction-limited beam quality. *Optics Express*, 24(4):4187, feb 2016.
- [84] Craig Robin, Iyad Dajani, and Benjamin Pulford. Modal instability-suppressing, single-frequency photonic crystal fiber amplifier with 811 w output power. *Optics Letters*, 39(3):666, jan 2014.
- [85] Clément Dixneuf, Germain Guiraud, Yves-Vincent Bardin, Quentin Rosa, Mathieu Goeppner, Adèle Hilico, Christophe Pierre, Johan Boullet, Nicholas Traynor, and Giorgio Santarelli. Ultra-low intensity noise, all fiber 365 w linearly polarized single frequency laser at 1064 nm. *Optics Express*, 28(8):10960, mar 2020.
- [86] L. Winkelmann, O. Puncken, R. Kluzik, C. Veltkamp, P. Kwee, J. Poeld, C. Bogan, B. Willke, M. Frede, J. Neumann, P. Wessels, and D. Kracht.

- Injection-locked single-frequency laser with an output power of 220 w. *Applied Physics B*, 102(3):529–538, feb 2011.
- [87] Felix Wellmann, Michael Steinke, Fabian Meylahn, Nina Bode, Benno Willke, Ludger Overmeyer, Jörg Neumann, and Dietmar Kracht. High power, single-frequency, monolithic fiber amplifier for the next generation of gravitational wave detectors. *Optics Express*, 27(20):28523, sep 2019.
- [88] Aaron Buikema, Franklin Jose, Steven J. Augst, Peter Fritschel, and Nergis Mavalvala. Narrow-linewidth fiber amplifier for gravitational-wave detectors. *Optics Letters*, 44(15):3833, jul 2019.
- [89] Nina Bode, Fabian Meylahn, and Benno Willke. Sequential high power laser amplifiers for gravitational wave detection. *Optics Express*, 28(20):29469, sep 2020.
- [90] Fabian Thies, Nina Bode, Patrick Oppermann, Maik Frede, Bastian Schulz, and Benno Willke. Nd:YVO4 high-power master oscillator power amplifier laser system for second-generation gravitational wave detectors. *Optics Letters*, 44(3):719, jan 2019.
- [91] P. Kwee, C. Bogan, K. Danzmann, M. Frede, H. Kim, P. King, J. Pödl, O. Puncken, R. L. Savage, F. Seifert, P. Wessels, L. Winkelmann, and B. Willke. Stabilized high-power laser system for the gravitational wave detector advanced LIGO. *Optics Express*, 20(10):10617, apr 2012.
- [92] the LIGO Scientific Collaboration. Instrument science white paper 2019. LIGO internal document LIGO-T1900409–v3.

- [93] Andrew P. Spencer. *Advanced Techniques in Laser Interferometry for Current and Future Gravitational Wave Detectors*. PhD thesis, University of Glasgow, August 2019.
- [94] Thomas J. Kane and Robert L. Byer. Monolithic, unidirectional single-mode nd:YAG ring laser. *Optics Letters*, 10(2):65, feb 1985.
- [95] I Freitag, A Tünnermann, and H Welling. Power scaling of diode-pumped monolithic nd:YAG lasers to output powers of several watts. *Optics Communications*, 115(5-6):511–515, apr 1995.
- [96] Michèle Heurs, Volker M. Quetschke, Benno Willke, Karsten Danzmann, and Ingo Freitag. Simultaneously suppressing frequency and intensity noise in a nd:YAG nonplanar ring oscillator by means of the current-lock technique. *Optics Letters*, 29(18):2148, sep 2004.
- [97] J. Lui, R. Savage, P. King, L. Zhang, and S. et al. Appert. aLIGO all-bolted PMC summary. LIGO Document T1700543-v2 <https://dcc.ligo.org/LIGO-T1700543>.
- [98] Patrick Kwee, Frank Seifert, Benno Willke, and Karsten Danzmann. Laser beam quality and pointing measurement with an optical resonator. *Review of Scientific Instruments*, 78(7):073103, jul 2007.
- [99] P. Kwee. *Laser Characterization and Stabilization for Precision Interferometry*. PhD thesis, Max Planck Institute for Gravitational Physics, 2010.

- [100] R. S. Abbott and P. J. King. Diode-pumped nd:YAG laser intensity noise suppression using a current shunt. *Review of Scientific Instruments*, 72(2):1346, 2001.
- [101] B W Barr, K A Strain, and C J Killow. Laser amplitude stabilization for advanced interferometric gravitational wave detectors. *Classical and Quantum Gravity*, 22(20):4279–4283, oct 2005.
- [102] Jameson Rollins, David Ottaway, Michael Zucker, Rainer Weiss, and Richard Abbott. Solid-state laser intensity stabilization at the 10^{-8} level. *Optics Letters*, 29(16):1876, aug 2004.
- [103] I Zawischa, M Brendel, K Danzmann, C Fallnich, M Heurs, S Nagano, V Quetschke, H Welling, and B Willke. The GEO 600 laser system. *Classical and Quantum Gravity*, 19(7):1775–1781, mar 2002.
- [104] F Nocera. LIGO laser intensity noise suppression. *Classical and Quantum Gravity*, 21(5):S481–S485, feb 2004.
- [105] P. Horowitz and W. Hill. *The Art of Electronics*. Cambridge University Pr., 2015.
- [106] P. Kwee. aLIGO PSL ISS inner-loop photodiode. LIGO Document D1001998-v2 <https://dcc.ligo.org/LIGO-D1001998/public>.
- [107] C. Bogan. Intensity Stabilization Servo Module (inner loop). LIGO Document D1001985-v2 <https://dcc.ligo.org/LIGO-D1001985/public>.
- [108] G. Mueller. Pointing Requirements for Advanced LIGO. LIGO Document T0900142-v2 <https://dcc.ligo.org/LIGO-T0900142>.

- [109] Euan Morrison, Brian J. Meers, David I. Robertson, and Henry Ward. Automatic alignment of optical interferometers. *Applied Optics*, 33(22):5041, aug 1994.
- [110] H. Kim, P. King, C. Krämer, P. Kwee, J. Pöld, R. Savage, P. Weßels, and B. Willke. Psl final design. DCC LIGO T0900649-v4, February 2010.
- [111] N. Uehara and K. Ueda. Accurate measurement of the radius of curvature of a concave mirror and the power dependence in a high-finesse fabry-perot interferometer. *Applied Optics*, 34(25):5611, sep 1995.
- [112] V. Frolov. DARM Optical Gain and Losses. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=42231>.
- [113] L. McCuller. Frequency Dependence of Squeezing Loss and Angle, Power Measurement by Radiation Pressure. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=52718>.
- [114] L. McCuller. Frequency Dependent Squeezing effects in the LIGO Intererometers. LIGO Document G2000863-v2 <https://dcc.ligo.org/LIGO-G2000863>.
- [115] V. Frolov. Arm Cavity Power, Power Recycling Gain, IFO Reflectivity and Arm Cavity Losses. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=42639>.
- [116] J. Richardson, M. Tse, and C. Austin. Measurement of SRC Gouy Phase. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=39779>.
- [117] C. Blair. OMC Scans. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=37742>.

- [118] C. Blair and V. Frolov. 40W OMC Scans Method and Preliminary Data. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=45830>.
- [119] M. Kasprzack, A. Effler, A. Mullavey, and M. Tse. HAM6 Work - M12 Move. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=37291>.
- [120] A. Effler and M. Kasprzack. LLO Interferometer Output Beam Study. LIGO Document T1800089-v2 <https://dcc.ligo.org/LIGO-T1800089>.
- [121] C. Blair, V. Frolov, and A. Effler. SRM CO2 Chop. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=38689>.
- [122] C. Blair. Annular CO2 1W at Various IFO Power. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=43065>.
- [123] C. Blair, V. Frolov, and A. Effler. OMC Mode Matching in Single Bounce. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=38698>.
- [124] A. Effler, V. Frolov, and C. Blair. OMC Scans of Cooling Interferometer. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=41428>.
- [125] A. Green, A. Effler, P. Dupej, V. Frolov, P. Fulda, and H. Yu. Finesse Squeezing Mode Matching Study for LLO. LIGO Document T1900159-v1 <https://dcc.ligo.org/LIGO-T1900159>.
- [126] A. Effler, P. Dupej, H. Yu, A. Green, and C. Blair. SQZ SRM Bounce 98% Mode Matching. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=42187>.
- [127] Daniel David Brown and Andreas Freise. Pykat, July 2017. <http://www.gwoptics.org/pykat>.

- [128] C. Blair. PRC Gouy phase measurement. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=49409>.
- [129] E. Knyazev, S. Danilishin, S. Hild, and F.Ya. Khalili. Speedmeter scheme for gravitational-wave detectors based on EPR quantum entanglement. *Physics Letters A*, 382(33):2219–2225, aug 2018.
- [130] Tobias Eberle, Sebastian Steinlechner, Jöran Bauchrowitz, Vitus Händchen, Henning Vahlbruch, Moritz Mehmet, Helge Müller-Ebhardt, and Roman Schnabel. Quantum enhancement of the zero-area sagnac interferometer topology for gravitational wave detection. *Physical Review Letters*, 104(25), jun 2010.
- [131] M S Stefszky, C M Mow-Lowry, S S Y Chua, D A Shaddock, B C Buchler, H Vahlbruch, A Khalaidovski, R Schnabel, P K Lam, and D E McClelland. Balanced homodyne detection of optical quantum states at audio-band frequencies and below. *Classical and Quantum Gravity*, 29(14):145015, jun 2012.
- [132] Daniel Sigg and. Status of the LIGO detectors. *Classical and Quantum Gravity*, 25(11):114041, may 2008.
- [133] Brian J. Meers and Kenneth A. Strain. Modulation, signal, and quantum noise in interferometers. *Physical Review A*, 44(7):4693–4703, oct 1991.
- [134] Sebastian Steinlechner, Bryan W. Barr, Angus S. Bell, Stefan L. Danilishin, Andreas Gläfke, Christian Gräf, Jan-Simon Hennig, E. Alasdair Houston, Sabina H. Huttner, Sean S. Leavey, Daniela Pascucci, Borja Sorazu, Andrew Spencer, Kenneth A. Strain, Jennifer Wright, and Stefan Hild. Local-oscillator noise coupling in balanced homodyne readout for advanced gravitational wave detectors. *Physical Review D*, 92(7), oct 2015.

- [135] Peter Fritschel, Matthew Evans, and Valery Frolov. Balanced homodyne readout for quantum limited gravitational wave detectors. *Optics Express*, 22(4):4224, feb 2014.
- [136] D.E. McClelland, N. Mavalvala, Y. Chen, and R. Schnabel. Advanced interferometry, quantum optics and optomechanics in gravitational wave detectors. *Laser & Photonics Reviews*, 5(5):677–696, mar 2011.
- [137] The LIGO Scientific Collaboration. A gravitational wave observatory operating beyond the quantum shot-noise limit. *Nature Physics*, 7(12):962–965, sep 2011.
- [138] Kiwamu Izumi and Daniel Sigg. Advanced LIGO: length sensing and control in a dual recycled interferometric gravitational wave antenna. *Classical and Quantum Gravity*, 34(1):015001, dec 2016.
- [139] S. Dwyer. O3b noise budget. <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=55755>.
- [140] S. Dwyer. personal communication.
- [141] B. Lantz. Optical Integration of the aLIGO isolation tables. LIGO Document G2001539-v2 <https://dcc.ligo.org/G2001539>.
- [142] T. Zhang. *Advanced techniques for future generation gravitational wave detectors*. PhD thesis, University of Glasgow, 2019.
- [143] Hang Yu Peter Fritschel, Vivishek Sudhir. LIGO DCC Document G1702485-v1.

- [144] P. Fritschel and S. Hild. Requirements for the HAM Relay Triple Suspension (HRTS). LIGO Document T1800413-v1 <https://dcc.ligo.org/LIGO-T1900817>.
- [145] C. Bond, P. Fulda, A. Freise, D. Brown, K. Kokeyama, L. Carbone, and A. Perreca. Finesse Input Files for the L1 Interferometer. LIGO Document T1300901-v5 <https://dcc.ligo.org/LIGO-T1300901>.
- [146] B. Lantz, R. Mittleman, J. Kissel, J. Warner, and A. Pele. Typical Motion of HAM Suspension Points in O2. LIGO Document T1800066-v2 <https://dcc.ligo.org/T1800066>.
- [147] A. Huddart. A+ HRTS Preliminary Design Document. LIGO DCC, February 2019. LIGO Document T1900817-v5 <https://dcc.ligo.org/LIGO-T1900817>.
- [148] L. Barsotti. aLIGO ISC Anti-symmetric Port: Beam Parameters and Optical Layout. LIGO Document T1200410-v2 <https://dcc.ligo.org/LIGO-T1200410/public>.
- [149] Peter Fritschel Jeff Lewis Sam Waldman Koji Arai, Sam Barnum. LIGO-T1000276-v5.
- [150] A. Brookes, P. Fritschel, F. Matichard, L. McCuller, M. Noh, and C. Torrie. Sams design requirements document. LIGO DCC, February 2019. E1800314.
- [151] Evgenia Kochkina. *Stigmatic and Astigmatic Gaussian Beams in Fundamental Mode: Impact of Beam Model Choice on Interferometric Pathlength Signal Estimates*. PhD thesis, Hannover U., 2013.

- [152] J. Briggs, M. Barton, J. Jones, E. Oelker, S. Webster, and K.A. Strain. A+ BHD ISC Additional Design Requirements. LIGO Document E2000072-v2 <https://dcc.ligo.org/LIGO-E2000072-v2>.
- [153] Ryan Lawrence, Michael Zucker, Peter Fritschel, Phil Marfuta, and David Shoemaker. Adaptive thermal compensation of test masses in advanced LIGO. *Classical and Quantum Gravity*, 19(7):1803–1812, mar 2002.
- [154] C. Blair. personal communication.
- [155] P. Fulda, R. Goetz, A. Green, R. Martin, and D. B. Tanner. A+ Output Faraday Isolator Final Design. LIGO Document T1900788-v2 <https://dcc.ligo.org/LIGO-T1900788>.
- [156] F. Bayer-Helms. Coupling coefficients of an incident wave and the modes of a spherical optical resonator in the case of mismatching and misalignment. *Applied Optics*, 23(9):1369, may 1984.
- [157] Antonio Perreca, Aidan F. Brooks, Jonathan W. Richardson, Daniel Töyrä, and Rory Smith. Analysis and visualization of the output mode-matching requirements for squeezing in advanced LIGO and future gravitational wave detectors. *Physical Review D*, 101(10), may 2020.
- [158] A.E. Siegman. *lasers*. University Science Books, 1986.
- [159] A.J. Moseley, M.D. Scott, A.H. Moore, and R.H. Wallis. High-efficiency, low-leakage MOCVD-grown GaInAs/AlInAs heterojunction photodiodes for detection to $2.4\mu\text{m}$. *Electronics Letters*, 22(22):1206, 1986.

- [160] Marius Grundmann. *The Physics of Semiconductors*. Springer Berlin Heidelberg, 2010.
- [161] Vladimir Protopopov. *Practical Opto-Electronics*. Springer International Publishing, 2014.
- [162] B. Lantz, S. L. Danilishin, S. Hild, Gustafson E., D. Coyne, V. Quetschke, G. D. Hammond, R. Adhikari, M. Evans, and R. Bassiri. Instrument science white paper 2018. LIGO DCC.
- [163] The LIGO Virgo Collaboration. Exploring the sensitivity of next generation gravitational wave detectors. *Classical and Quantum Gravity*, 34(4):044001, jan 2017.
- [164] Georgia L. Mansell, Terry G. McRae, Paul A. Altin, Min Jet Yap, Robert L. Ward, Bram J. J. Slagmolen, Daniel A. Shaddock, and David E. McClelland. Observation of squeezed light in the $2\mu\text{m}$ region. *Physical Review Letters*, 120(20), may 2018.
- [165] M. J. Yap, D. W. Gould, T. G. McRae, P. A. Altin, N. Kijbunchoo, G. L. Mansell, R. L. Ward, D. A. Shaddock, B. J. J. Slagmolen, and D. E. McClelland. Squeezed vacuum phase control at $2\mu\text{m}$. *Optics Letters*, 44(21):5386, nov 2019.
- [166] M. Gendry, V. Drouot, C. Santinelli, and G. Hollinger. Critical thicknesses of highly strained InGaAs layers grown on InP by molecular beam epitaxy. *Applied Physics Letters*, 60(18):2249–2251, may 1992.

- [167] Ramon U. Martinelli, Thomas J. Zamerowski, and Paul A. Longeway. 2.6 μm InGaAs photodiodes. *Applied Physics Letters*, 53(11):989–991, sep 1988.
- [168] Lars Zimmermann, Joachim John, Stefan Degroote, Gustaaf Borghs, Chris Van Hoof, and Stefan Nemeth. Extended wavelength InGaAs on GaAs using InAlAs buffer for back-side-illuminated short-wave infrared detectors. *Applied Physics Letters*, 82(17):2838–2840, apr 2003.
- [169] Gregory M Harry and the LIGO Scientific Collaboration. Advanced LIGO: the next generation of gravitational wave detectors. *Classical and Quantum Gravity*, 27(8):084006, apr 2010.
- [170] Teng Zhang, Denis Martynov, Andreas Freise, and Haixing Miao. Quantum squeezing schemes for heterodyne readout. *Physical Review D*, 101(12), jun 2020.
- [171] A. A. Muzammil, C. Mueller, and G. Mueller. Optical Layout and Parameters for the Advanced LIGO Cavities. LIGO Document T0900043-v11 <https://dcc.ligo.org/LIGO-T0900043/public>.
- [172] LIGO Optics Database. <https://galaxy.ligo.caltech.edu/optics/>.
- [173] R. Martin and G. Billingsley. Measurement Uncertainties in Determining the Radius of Curvature for SR2-04. LIGO Document T1300653-v1 <https://dcc.ligo.org/LIGO-T1300653>.
- [174] G. Billingsley. SR301 Figure Measurement. LIGO Document E1101231-v1 <https://dcc.ligo.org/E1101231-v1>.

- [175] M. Smith and D. Coyne. Advanced LIGO L1 Optical Layout, ZEMAX. LIGO Document D0902216-v8 <https://dcc.ligo.org/LIGO-D0902216>.